# Lecture 1 - Class Overview and Algebra PreCalculus Review

# Lecture 1 - Class Overview and Algebra PreCalculus Review

Garrett Credi

Summer 2026

# Welcome to MATH 1271!




Figure: Our Mascot, Maxwell, Deep in Thought

# Today's Goals

1. Overview of the Course
2. Lecture on Algebra/Precalc Background
3. Worksheet Time!

# Course Policies I

I've structured this course in the following manner:

1. We will start class with a *5 minute* quiz on the topic(s) covered in the previous lecture
2. We will then begin an  75 minute lecture, with a  5-ish minute break about halfway through
3. After another short break I'll hand out worksheets where we'll get a chance to apply the concepts we learned during lecture

# Course Policies II

After each lecture you will have some *homework* due at 11:59 PM the following day. It will:

1. Be graded on an *honest completion* basis
2. Be split up into certain “levels”
3. And will conclude with one “challenge problem” designed to help you all develop more confidence for quiz/exam environments

# Course Policies III

We will also have *weekly engagement portfolios* which are short  100 word pieces of writing to get you to engage in some metacognition about how you are learning in this course.
These will be consistently due at 11:59 PM on Mondays.

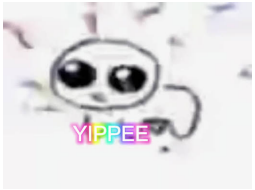
# Course Policies IV

As mentioned before, we will have quizzes at the start of class relating to the content included in the previously-due homework, graded for accuracy.
We will also have *three* in class exams:

1. Exam 1: 10/6
2. Exam 2: 11/3
3. Exam 3: 12/8

Which will all take place during our regular class meeting here (Tate Hall 120).
Finally we will have a *cumulative* final exam on 12/17 from 12PM-3PM Location TBD.

# Course Policies V




Some things that might make you happy!

- (At this point) your two lowest quizzes and two lowest engagement portfolios will be dropped.
- If you do poorly on an exam, you can have your performance on the final exam replace the grade if it is better.
- All this information is easily accessible and searchable on our syllabus on Canvas!

# Office Hours?

I'm planning on holding two hours of office hours a week: in order that I hold them at the most opportune times please indicate your desires/availability on the When2Meet here.
There are also Office Hours that are held by some other TAs that you can find here.

# Who Am I?

Hi! I'm Garrett! I am:

1. A 3rd Year PhD student in the math department
   (1) My research is in a field of math called *algebraic topology*
2. Someone who has TA'd for math courses three times (1x1271, 2x3283W) and taught once (1x2263)
3. The proud cat dad of the lovely, gracious, cute, and venerable Maxwell (who is our course banner on canvas!)
4. Super excited to teach you all some calculus this semester!!!

You can contact me at credi008@umn.edu

# Who are you?

What is your:

1. Name
2. Major and standing (freshman, sophomore, etc)
3. Favorite *thing*?

# Let's Do Some Math!

I want to spend today making sure we're all clear on the mathematical background necessary for success in this class:

1. Functions, their equations, graphs, and properties (e.g. domain and range)
2. Techniques with fractions
3. Trigonometric functions
4. Solving Equations

# Functions I

What is a function?

## Function

A function is an association from "things" of type $A$ to other "things" of type $B$, typically denoted as $f : A \to B$.
For the purposes of our class, we will study **real-valued** functions of a **real variable** e.g. functions that take in a real number $x \in \mathbb{R}$ (read as $x$ is in the real numbers) and output a real number $y \in \mathbb{R}$.

## Equation of a Function

Real valued functions of a single real variable (we'll just call these functions from now on) can be **defined** by an equation which is just an algebraic specification of what the function does to an input number.

E.g. we can define the *squaring* function as $f(x) = x^2$. This means the value of the function $f$ at some real number $x$ is the square of $x$, so, for instance, $f(2) = 2^2 = 4, f(3) = 3^2 = 9$, etc.

# Functions II

## Graph of a function

The **graph** of a function $f : \mathbb{R} \to \mathbb{R}$ is a graphical/geometric way of interpreting what a function does.
To draw the graph, we first draw a *2D-Coordinate Grid*, and then draw all the points $(x, f(x))$ connected via a smooth line.

E.g.

# Functions III

Sometimes functions behave differently depending on what kind of real number we input into them for instance the *absolute value function* leaves positive real numbers unchanged ($|2| = 2$) but turns negative numbers into their opposites ($|-2| = 2$). How can we define these functions?

## Piecewise Function

A function is **defined piecewise** by the following method:

$$f(x) = \begin{cases} f_1(x), & \text{some condition on x} \\ f_2(x), & \text{some other condition on x} \\ f_3(x), & \text{maybe some other condition on x} \\ \vdots & \\ g(x), & \text{otherwise if none of the above conditions are met} \end{cases} \tag{1}$$

# Functions IV

For example, we define the absolute value function as:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0 \end{cases} \tag{2}$$

and we can define some wacky function as:

$$wacky(x) = \begin{cases} x^2, & x > -5 \\ -x^3 + 2x + 1, & -5 \leq x < 2 \\ 1, & 2 \leq x < 3 \\ \frac{x}{2} + 3, & \text{otherwise} \end{cases} \tag{3}$$

# Functions V

There are two important properties of a function that we need to talk about: domain and range.

## Domain and Range of a Function

The domain of a function is the set of all values the function accepts as input. It is kind of arbitrary! However, we'll follow the convention that the domain should be the **largest** set of values it could accept as input.
Similarly, the range of a function is the set of all values the function could possibly output over all possible inputs. Given a specified domain, there is **exactly one** set of ranges so we don't have to deal with any arbitrariness.

For example, the *square-root* function $f(x) = \sqrt{x}$ would have:

- Domain: the *non-negative real numbers* $[0, \infty)$
- Range: again the *non-negative real numbers* $[0, \infty)$

What about a function like $f(x) = x^2 + 1$?

# Fractions I

It's also worth going over some fraction foundations which will be very useful very soon in the course:

## Addition of Fractions

## Multiplication of Fractions

# Fractions II

I also want to make clear what people mean by:

- Clearing out common factors/
- Factoring [blah] out/
- Multiplying by 1

**The Multiplying by 1 Trick**

# Special Functions

We have a class of “special” functions:

- The exponential function $\exp(x) = e^x$
- It’s inverse, the natural logarithm $\ln(x)$
- The power functions $x^n$
- Their inverses $\sqrt[n]{x}$
- The trigonometric functions

# Trigonometric Functions

# Solving Equations I

Oftentimes we're interested in finding points where functions attain certain values or exhibit important behavior, how do we find them?
For instance, let's say we want to buy some ramen and the cost of ramen in dollars varies by weight in pounds according to the following function:

$$P(w) = x^2 + 7x.$$

If we only have 30 dollars, how much ramen can we buy?

# Solving Equations II

How do we set that up mathematically?
We want to find the weight $w$ so that:

$$P(w) = \$30 \tag{4}$$

So we can use our definition of $P$ to get:

$$\begin{aligned} w^2 + 7w &= 30 \\ w^2 + 7w - 30 &= 0 \\ (w + 10)(w - 3) &= 0 \\ w = -10 \text{ or } w &= 3 \end{aligned}$$

And since $-10$ lbs of ramen doesn't make physical sense, we get that we can by 3 lbs of ramen with that much money.

# Solving Equations III

Why did we choose to solve

$$(w + 10)(w - 3) = 0$$

for $w = -10, w = 3$
instead of solving

$$w(w + 7) = 30$$

for $w = 30$ or $w + 7 = 30$?

**No Zero Divisors**

If there are two real numbers $a, b$ so that $ab = 0$, it **MUST** be the case that $a = 0$ or $b = 0$.

Question: If there are two real numbers $a, b$ so that $ab = 30$, must it be the case that $a = 30$ or $b = 30$?

# Solving Equations IV

How might we approach solving for $x$ in an equation like:

$$(e^x)^2 - 5e^x + 6 = 0? \tag{5}$$

Or maybe like:

$$2\sin(x)^2 + \cos(x)^2 = 2? \tag{6}$$

# Worksheet Time

Let’s get into groups and work on the worksheet!

# End Of Class

That’s all folks!

# Lecture 2 - Intro To Limits

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# Day 1 Recap

Questions:

1. Was everyone able to access the Gradescope/homework?
2. How did the homework go? (Note: Length will not be as long for future HWs, this one just had a lot of topics to review)
3. Any questions, comments, concerns?

# Quiz Time!

At this point it's quiz time!

1. Get into neatly organized rows
2. Do not flip over the page and start until directed to
3. Should go without saying but no cheating!
   - No looking at neighbors papers
   - No calculators
   - No notes
   - No phones
   - etc. etc.
4. I'll write on the board the minutes you have left, crossing each number out when you have *less than* that number of minutes left to complete
5. When time is up, stop writing and hold your quiz up so I can grab it. If you finish early, just chill until the 5 mins are up.

# Time to do some math!

Now let’s start the lecture!

# MATH 1271 - Part 1: Limits

We will start our course by discussing the concept of a *limit of a function*.
Intuitively, the limit serves several important purposes:

- It gives you information about the value that a function tends towards, as it's input tends towards some fixed input.
- It tells you what value the function is "approximating" around some fixed input.
- It tells you local information about the function around some *small neighborhood* of some fixed input.

# Approximation I

Let's run with the second interpretation for a bit:
Say you're doing some experiment where you're studying the relationship between some *input variable x* (maybe it's the concentration of some acid in a solution) and some *output variable y* (maybe it's the pH of that solution).
Real world problems:

1. We can almost never perfectly measure physical quantities e.g.
   (1) We can't know the exact number of molecules of an acid present in a solution, only up to some error
   (2) Using typical measurements like pH strips etc. we also can't get a precise measurement of the output variable either

How do we proceed if, for instance, we want to know the *exact* pH of a solution whose concentration of acid is *exactly* 0.5 moles per liter?

# Approximation II

We can either:

1. Make many many many measurements and take their average, or
2. Use more precise methods of measurements that decrease the error range as much as possible.

Graphically:

# Approximation III

**WARNING!**

How do we actually know that this average we're taking is the true output value?

What it really is, is...
The Limit!

# Limits I

## Limit of a Function

For some function $f : \mathbb{R} \to \mathbb{R}$ and some fixed input $x_0 \in \mathbb{R}$, we say that $\lim_{x \to x_0} f(x) = L$ (said "the limit of $f(x)$ as $x$ approaches $x_0$ is $L$") if we can make $f(x)$ as close to $L$ as we want just by forcing $x$ to be close enough to $x_0$.
Explicitly: for any error tolerance $\epsilon > 0$, if we want $f(x)$ to be equal to $L$ within an error of $\epsilon$ (e.g. $|f(x) - L| < \epsilon$), then we just need to find some input error tolerance $\delta > 0$ and ensure that $x$ is equal to $x_0$ within an error of $\delta$ (e.g. $|x - x_0| < \delta$).
If no such value $L$ exists (i.e. the function *cannot* be well approximated around $x_0$) then we say that $\lim_{x \to x_0} f(x)$ *does not exist*.

# Limits II

How can we find limits?
Option 1 (easiest): Graphically!
Given a graph of a function, we can easily tell what value a function approaches just by examining where the graph tends towards as we near a given point.

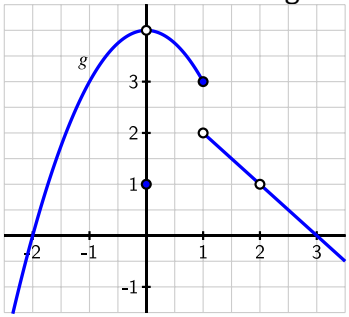



Figure: An example function

# Limits III

What if we don't have a graph to work with?

Option 2 (medium easy, *only an estimate!*): Evaluation!

Given an expression of a function, e.g. $f(x) = x^2$ and a specific value around which to compute the limit, e.g. $x_0 = 2$, we can compute values that are closer and closer to $x_0$ *by hand/calculator* and guess if they're approaching a value. (Similar to the method we talked about in the approximation section.)

# Limits IV

What about if we are both:

1. Not given a graph
2. Interested in the *exact* limit, not an approximation?

Option 3 (exact!): Algebraic Manipulation!
E.g., if we wanted to find $\lim_{x \to 2} x^2$ we could:

1. Rewrite it as $\lim_{h \to 0} (2+h)^2$
2. Expand the argument to get $\lim_{h \to 0} 4 + 2h + h^2$
3. Group the $h$ terms together to get $\lim_{h \to 0} 4 + h(2+h)$
4. Notice that as $h \to 0$, $h(2+h) \to 0$ as well so $\lim_{x \to 2} x^2 = 4$

# Limits V

Notice that $\lim_{x \to 2} x^2 = 4 = 2^2 = f(2)$ i.e. *the limit of* $f(x)$ *as* $x \to x_0$ *is equal to* $f(x_0)$!
Is this always true? I.e. no matter what function $f$ we choose and what fixed input $x_0$ we take, do we always have that:

$$\lim_{x \to x_0} f(x) \stackrel{?}{=} f(x_0) \tag{7}$$

## NO!

Two things can go wrong:

1. We can artifically redefine the value of a function as much as we want:

$$f(x) = \begin{cases} 4, & x = 0 \\ x^2, & x \neq 0 \end{cases} \tag{8}$$

2. We can have a function that isn't *defined* at a point but still *has a limit at that point*:

$$g(x) = \frac{x^2 - 1}{x + 1}. \tag{9}$$

# Limits VII

Let's study that $g(x) = \dfrac{x^2 - 1}{x + 1}$ function more:

1. What is the domain of $g$?
2. For points $x_0$ in the domain of $g$, is it true that $\lim_{x \to x_0} g(x) = g(x_0)$?
3. What about the point that is *not* in the domain of $g$? What is $\lim_{x \to -1} g(x)$?

# Limits VIII

Let's try another weird function:

$$h(x) = \sin(\frac{\pi}{x}). \tag{10}$$

1. Domain?
2. Does limit=value for points in the domain?
3. What about $x_0 = 0$?

# Instantaneous Velocity I

What can we *do* with limits?

## Average Velocity

Suppose that an object's position as a function of time $t$ is given by the function $s(t)$.
The average velocity of the object, from time $t = a$ to time $t = b$ is given by:

$$AV_{[a,b]} := \frac{s(b) - s(a)}{b - a}. \tag{11}$$

However, we rarely ever see *average* velocity in real life! E.g. if you drive a car then the dashboard reports your speed exactly in the instant you look at it, not over the last 1, 2, 5, etc. minutes!
(If it did, imagine someone driving slowly for a while, rapidly accelerating to a dangerous speed, getting pulled over and then trying to tell the cop "oh well my car said my average speed over the past 5 minutes was just 20mph" unlikely to avoid a ticket!)

# Instantaneous Velocity II

That why we care about the *instantaneous* velocity of an object:

## Instantaneous Velocity

Suppose that an object's position as a function of time $t$ is given by the function $s(t)$. The instantaneous velocity of the object at time $t = a$ is defined as

$$IV_{t=a} = \lim_{\Delta t \to 0} AV_{[a,a+\Delta t]} = \lim_{h \to 0} \frac{s(a+h) - s(a)}{h} \tag{12}$$

# Instantaneous Velocity III

Suppose that $s(t) = \sqrt{t+1}$. What is the instantaneous velocity at $t = 0$?

1. What is the domain of $\dfrac{s(0+h) - s(0)}{h} = \dfrac{\sqrt{1+h} - 1}{h}$?
2. Can we determine the limit just by evaluating the above function at $h = 0$?
3. How could we get $\dfrac{\sqrt{1+h} - 1}{h}$ into a form where we *could* just evaluate?

# Worksheet Time!

Now let's get into groups and work on the worksheet!

# End of Class!

That’s all folks!

# Lecture 3 - Limits II

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# Announcements

Homework Policy:

- For homework assigned on Tueday, the due date will be Thursday @ 6PM
- For homework assigned on Thursday, the due date will be Saturday @ 6PM

Office Hours:

- Wednesday 12:15 - 1:15 PM
- Thursday 4:30 - 5:30 PM

Both are in Vincent Hall 105.

# Last Time:

Last lecture we:

1. Defined the notion of a *limit* of a function: $\lim_{x \to a} f(x) = L \in \mathbb{R}$ if we can make $f(x) \approx L$ up to some error ($\epsilon > 0$) just by forcing $x \approx a$ up to some error ($\delta > 0$)
2. Said that if no such value $L$ existed we said that $\lim_{x \to a} f(x)$ *did not exist* D.N.E.
3. Computed limits graphically:
   - points where the graph of a function "broke" were points where the limit did not exist
   - points where the graph of a function had a "hole" had the limit exist but it ***was not equal to the function's value***
   - points where the graph was just an unbroken curve had the limit exist and *equal* to the value at that point
4. *Approximated* limits using a table of values
5. Computed limits by algebraic manipulation

# Today's Goals:

Today we will develop our ability to algebraically compute limits by:

1. Studying the "simplest" functions to take limits of, *continuous functions*
2. Discovering *limit properties* that help us reduce complicated limits to easier limits
3. Combining the above to get the process of algebraically computing limits

And we will also study cases where the limit is infinite.

# Continuous Functions I

To start off: let's consider the function:

$$f(x) = \frac{x^3 - x^2 - x + 1}{x^2 + x - 2} \tag{13}$$

1. What is the domain of $f$?
2. Using the graph of $f(x)$, what are:
   (1) $\lim_{x \to 3} f(x)$?
   (2) $\lim_{x \to 5} f(x)$?
   (3) $\lim_{x \to 1} f(x)$?

# Continuous Functions II

Those first two points, where we have

$$f(3) = \lim_{x \to 3} f(x) \quad f(5) = \lim_{x \to 5} f(x)$$

are a special kind of point:

**Continuity at a point**

Given a function $f(x)$ and a real number $a$ , we say that $f(x)$ *is continuous at* $a$ if:

1. The limit $\lim_{x \to a} f(x)$ exists
2. $a$ is in the domain of $f$
3. $\lim_{x \to a} f(x) = f(a)$

# Continuous Functions III

We can define classes of functions that are specified to be continuous at important collections of points:

**Continuous on $[a, b]$**

A function $f(x)$ is said to be continuous on the interval $[a, b]$ if, for every real number $a \leq x_0 \leq b$ we have that $f(x)$ is continuous at $x_0$.

**Continuous Function**

A function $f(x)$ is said to be continuous if, for every real number $x_0$ in the domain of $f$, we have that $f(x)$ is continuous at $x_0$.

# Continuous Functions IV

Continuous functions are the “atoms” we want to consider when computing limits since we can replace the difficult task of *proving* what number a functions value approaches by the easier task of *evaluating* the function.
What are some continuous function?

## Important Continuous Functions

The following are examples of continuous functions:

1. The constant function $f_c(x) = c$ for some constant number $c$.
2. The identity function $f_{id}(x) = x$
3. The exponential and logarithm functions $\exp(x) = e^x, \ln(x)$
4. All trigonometric functions, $\sin(x), \cos(x), \tan(x)$...

# Continuous Functions V

What about other familiar functions like:

1. Power functions $x^n$,
2. Polynomials $c_0x^n + c_1x^{n-1} + \ldots + c_nx + c_{n+1}$,
3. Rational functions $\dfrac{p(x)}{q(x)}$
4. Weird functions like $e^{\sin(x)} + \cos(x^n)\sqrt{x}$?

We can determine the points at which these functions are continuous (and thus the points at which it is easy to calculate limits) by discovering *limit properties*!

# Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$
2. Property 2 says that as long as $x \approx 0$ then $g(x) \approx 0$

So if $x \approx 0$ then $h(x) = f(x) + g(x) \approx 1 + 0 = 1$!

# Limit Properties: Additivity II

Let's codify this into a property:

**Additivity of Limits**

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x) + g(x))$ exists and is equal to $b + c$.

# Limit Properties: Multiplicativity

## Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and $a$ is a real number so that

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $\lim_{x \to a} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a}(f(x)g(x))$ exists and is equal to $bc$.

Similarly:

## Limits of Multiplicative Inverses

Suppose that $f(x)$ is a function and $a$ is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number $b$, and that $b \neq 0$.

Then we have that $\lim_{x \to a} \dfrac{1}{f(x)}$ exists and is equal to $\dfrac{1}{b}$.

# Limit Properties: Composites

## Limits of Composites

Suppose that $f(x)$ and $g(x)$ are two functions, that $a$ is a real number, and that:

1. $\lim_{x \to a} f(x)$ exists and is equal to $b$
2. $f(x)$ is *not* constantly equal to $b$ around $x = a$,
3. $\lim_{x \to b} g(x)$ exists and is equal to $c$

Then we have that $\lim_{x \to a} g(f(x)) = c$.

# Combining Continuous Functions

We can use the above limit laws to enlarge the class of functions we know to be continuous!

## Combining Continuous Functions

Suppose that $f(x)$ and $g(x)$ are two functions that are continuous at some real number $a$. Then we have that:

1. $f(x) + g(x)$ is continuous at $x = a$
2. $f(x)g(x)$ is continuous at $x = a$
3. If $g(a)$ and $\lim_{x \to a} g(x)$ are both nonzero then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$.
4. If $g(x)$ is instead continuous at $f(a)$ then $g(f(x))$ is continuous at $x = a$.

# Moral of the Story

The big upshot of this is the following:

**Strategy for Computing Limits**

If you are tasked with computing a limit $\lim_{x \to a} F(x)$ algebraically your goal should be to reduce the expression of the functions down to end up with an equivalent limit $\lim_{x \to a} F(x) = \lim_{x \to a} f(x)$ where:

1. $f(x)$ is a continuous function
2. and $a$ is in the domain of $f(x)$.

# Moral of the Story: Example

For instance:

$$\lim_{x\to 1}\frac{x^3-x^2-x+1}{x^2+x-2}=\lim_{x\to 1}\frac{x^3-x^2-x-1}{(x-1)(x+2)} \tag{14}$$

$$=\lim_{x\to 1}\frac{(x-1)(x^2-1)}{(x-1)(x+2)} \tag{15}$$

$$=\lim_{x\to 1}\frac{(x-1)}{(x-1)}\frac{x^2-1}{x+2} \tag{16}$$

$$=\lim_{x\to 1}\frac{x^2-1}{x+2} \tag{17}$$

$$=\frac{(1)^2-1}{1+2}=\frac{1-1}{3}=\frac{0}{3}=0 \tag{18}$$

# One Sided Limits

We can also analyze a similar kind of quantitiy to the limit that will end up being useful when we study limits at infinity on Thursday:

## One-Sided Limits

Given a function $f(x)$ and a number $a$:

- We say that $\lim_{x \to a^-} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the left*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x < a$.
- Similarly, we say that $\lim_{x \to a^+} f(x) = L$ (verbally, "the limit of $f(x)$ as $x$ approaches $a$ *from the right*") if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x \approx a$ (up to some error $\delta > 0$) and $x > a$.

# Relationship between Limits and One-Sided Limits

Questions:

1. Can we have that $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$?
2. If $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$ what can we say about $\lim_{x \to a} f(x)$?
3. If $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$, what can we say about $\lim_{x \to a} f(x)$?
4. If $\lim_{x \to a^-} f(x) = f(a) = \lim_{x \to a^+} f(x)$, what previously discussed definition does the function $f(x)$ and the point $a$ satisfy?

# Worksheet Time!

Let’s get into groups and work on the worksheet!

# End of Class!

That’s all folks!

# Lecture 4 - Limits III

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# Quiz Time!

Let's get into formation for quiz time.

# Last Time

We spent much of last lecture developing our ability to compute limits via:

1. Recognizing when functions were *continuous* so that we could reduce the computation of limits to simpler evaluations
2. Developing limit properties which allowed us to build up the computation of complicated limits via reduction to simpler ones

We also developed the notion of the *one sided limits* - restricted kinds of limits where we need only approach from a singular side.

Today's Goals

We'll apply both our new limit computation skills as well as our knowledge of one-sided limits to work with *limits at or involving infinity*.
We will also uncover our last tool to help us compute limits: *the Squeeze Theorem*.

# Prelude I

Let's first examine one of the problems from the previous worksheet:
We had two functions whose graphs looked like:

And we wanted to compute $\lim_{x \to 1} g(f(x)) = \lim_{x \to 1} g((x-1)^2) = \lim_{u \to 0^+} g(u) = 3$

# Prelude II

Key to that problem was noticing that:

$$x \to 1 \quad \iff \quad (x-1) \to 0^{+} \tag{19}$$

Which allowed us to do a “substitution” that made the limit easier to compute.
This will be our strategy for computing limits at $\pm\infty$! Since $\pm\infty$ is not an actual number we can compute with in a consistent/coherent way, we’ll try and replace $x \to \pm\infty$ with a limit as *some other quantity* approaches *some actual number*.

# Limits at Infinity I

Before that, let's actually look at the literal definition to see why we want to avoid it:

**Limits at Infinity**

For a function $f(x)$, we say that $\lim_{x\to\infty} f(x) = L$ if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x > N$ for some real number $N$.
Similarly, we say that $\lim_{x\to-\infty} f(x) = L$ if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x < M$ for some real number $M$.

Do we want to prove things in this class like we would need to if we actually used this definition? NO!

# Limits at Infinity II

Instead, let's see if we can relate it to something we could instead *compute* using our tools from last lecture.

$$x > N \iff 0 > x > N \tag{20}$$
$$\iff 0 < \frac{1}{x} < \frac{1}{N} \tag{21}$$
$$\iff 0 < \frac{1}{x} - 0 < \frac{1}{N} \tag{22}$$

In other words:

**Limits at Infinity in terms of Finite Limits**

Taking the limit as $x \to +\infty$ is *exactly equivalent* to taking the limit as $\frac{1}{x} \to 0^+$

Similarly $x \to -\infty$ is *exactly equivalent* to $\frac{1}{x} \to 0^-$!

# Limits at Infinity III: Example

For example:

$$\lim_{x\to\infty}\frac{3x^2+2x+1}{2x^2+6x+131}=\lim_{\frac{1}{x}\to 0^+}\frac{3x^2+2x+1}{2x^2+6x+131} \tag{23}$$

$$=\lim_{u\to 0^+}\frac{3(\frac{1}{u})^2+2(\frac{1}{u})+1}{2(\frac{1}{u})^2+6(\frac{1}{u})+131} \tag{24}$$

$$=\lim_{u\to 0^+}\frac{u^2}{u^2}\frac{3\frac{1}{u^2}+2\frac{1}{u}+1}{2\frac{1}{u^2}+6\frac{1}{u}+131} \tag{25}$$

$$=\lim_{u\to 0^+}\frac{3+2u+u^2}{2+6u+131u^2} \tag{26}$$

$$=\lim_{u\to 0}\frac{3+2u+u^2}{2+6u+131u^2}=\frac{3+2\cdot 0+0^2}{2+6\cdot 0+131\cdot 0^2}=\frac{3}{2} \tag{27}$$

# Limits at Infinity IV: Example 1+2

So we took a scary looking limit $\lim_{x\to\infty} \frac{3x^2+2x+1}{2x^2+6x+131}$ and reduced it to simply evaluating some rational function $f(u) = \frac{3+2u+u^2}{2+6u+131u^2}$ at $u = 0$!
What about for just a polynomial?

$$\lim_{x\to-\infty}(x^3+2x+1) = \lim_{u\to0^-}((\frac{1}{u})^3+2\frac{1}{u}+1) \tag{28}$$

$$= \lim_{u\to0^-}\frac{1}{u^3}(1+2u^2+u^3) \tag{29}$$

$$\text{“}=\text{”}(\lim_{u\to0^-}\frac{1}{u^3})\cdot(\lim_{u\to0^-}(1+2u^2+u^3)) \tag{30}$$

$$= \lim_{u\to0^-}\frac{1}{u^3} \tag{31}$$

Where we unfortunately encounter a problem since $\frac{1}{u^3}$ has an *asymptote* at $u = 0$ and so cannot be replaced by a function continuous at 0. What do we do?

# Limits at Infinity V: Example 2 and Divergence to Infinity

Let's look at the graph!

## Limits Diverging to Infinity

For a function $f(x)$ and a point $a$, we say that $\lim_{x \to a} f(x) = +\infty$ (read as the limit of $f(x)$ as $x$ approaches $a$ diverges to positive infinity) if, for any lower bound $L$ we can force $f(x) > L$ just by forcing $x \approx a$ (up to some error $\delta > 0$).
Similarly, we say that $\lim_{x \to a} f(x) = -\infty$ (... diverges to minus infinity) if, for any upper bound $U$ we can force $f(x) < U$ just by forcing $x \approx a$ (up to some error $\delta > 0$).

So, from the graph of $\frac{1}{u^3}$ as $u \to 0^-$ we see that $\lim_{u \to 0^-} \frac{1}{u^3}$ *diverges* to minus infinity.

# Squeeze Theorem I

Let's consider one final limit:

$$\lim_{x\to\infty} \frac{\sin(x)}{x} = \lim_{u\to 0^+} \frac{\sin(\frac{1}{u})}{\frac{1}{u}} \tag{32}$$

$$= \lim_{u\to 0^+} u\sin(\frac{1}{u}) \tag{33}$$

And now we *REALLY* get stuck: as $u \to 0$, $\sin(\frac{1}{u})$ oscillates infinitely rapidly and escapes all ability to be pinned down to a value we could use to simplify the limit. Noooooo!
... ... but are we really stuck?

# Squeeze Theorem II

How much does $\sin(\frac{1}{u})$ really affect the value of the limit? Since we know that $\sin$ as a function never gets large at all, intuitively it shouldn't be able to outpace $u$ shrinking towards 0. Formally, since $-1 \leq \sin(x) \leq 1$ (the range of $\sin(x)$ is $[-1, 1]$) we have both that:

$$\lim_{u \to 0^+} u \sin(\frac{1}{u}) \leq \lim_{u \to 0^+} u \cdot 1 \tag{34}$$

$$= \lim_{u \to 0^+} u = \lim_{u \to 0} u \tag{35}$$

$$= 0 \tag{36}$$

and that:

$$\lim_{u \to 0^+} u \sin(\frac{1}{u}) \geq \lim_{u \to 0^+} u \cdot -1 \tag{37}$$

$$= -\lim_{u \to 0^+} u = -\lim_{u \to 0} u \tag{38}$$

$$= -0 = 0 \tag{39}$$

# Squeeze Theorem III

So if

$$0 \leq \lim_{u \to 0^+} u \sin(\frac{1}{u}) \leq 0 \tag{40}$$

we must have that $\lim_{u \to 0^+} u \sin(\frac{1}{u}) = 0$!
This style of argument comes from the following result:

## The Squeeze Theorem

Suppose we have a function $f(x)$ and a point $a$ so that:

1. we can find other functions $g(x), h(x)$ so that (around $x = a$) $g(x) \leq f(x) \leq h(x)$
2. and that $\lim_{x \to a} g(x) = \lim_{x \to a} h(x)$

Then we must have that $\lim_{x \to a} f(x)$ exists and is equal to the common value of $\lim_{x \to a} g(x) = \lim_{x \to a} h(x)$.

# Tools for Algebraically Computing Limits

Let's recap our tools:

1. Evaluation for continuous functions: if $f(x)$ is continuous at $x = a$ then

$$f(a) \quad = \quad \lim_{x \to a^+} f(x) = \lim_{x \to a} f(x) = \lim_{x \to a^-} f(x) \tag{41}$$

2. Reduction to continuous functions: maybe the point at which you are taking the limit towards isn't in the domain but you can replace it with one and use tool #1
3. Limit Properties: You can rewrite your limit as a sum, product, quotient, or composite of other limits and reduce it to computing those
4. Inversion for Limits at infinity: $x \to \infty \iff \frac{1}{x} \to 0^+$, $x \to -\infty \iff \frac{1}{x} \to 0^-$
5. Squeeze Theorem: maybe you have a limit which involves a term that "shouldn't matter" intuitively and you can use the range of that term to put bounds that squeeze the value of the limit.

# Worksheet Time

Let's get into groups and work on the worksheet!

# End of Class

That’s all folks!

# Lecture 5 - The Derivative

# Lecture 5 - The Derivative

Garrett Credi

Summer 2026

# Homework Updates/Info/Yappery?

After reading the engagement portfolios, I just had a few things I thought might be useful to say:

1. For choosing problems - I'd recommend looking over each of the problem types and choosing *a few problems from many problem types*.
2. Come review time (for quizzes and exams) the other problems you did not solve will be very useful practice problems (homework's will be uploaded on Canvas since evil gradescope doesn't let you see them after the due date)
3. If you're choosing homework problems in a way that's making you feel lost or making you feel unprepared for the quizzes please talk to me so we can discuss different strategies :)

# Quiz Time!

Let's get into nice orderly rows, spread out as much as possible, and slay this quiz :)

# The Road So Far

So far we've developed this notion of a limit:

1. Limits $\lim_{x \to a} f(x)$ are *the values that $f(x)$ approximates as $x$ approximates $a$ to closer and closer degrees*
2. Functions whose *actual* values match their approximated values at $x = a$ (e.g. where $f(a) = \lim_{x \to a} f(x)$) are called *continuous at $x = a$*. c.f. continuous on $[a, b]$, and continuous
3. We developed *maaaany* different tools to compute limits.

# Constant Value Approximation

For a function $f(x)$ that is *continuous at* $x = a$, how good is the approximation $f(x) \approx f(a)$ when $a \approx x$?

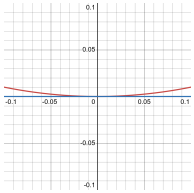



Figure: The graph of $f(x) = x^2$ near $x = 0$

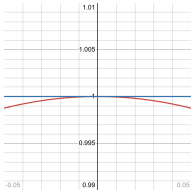



Figure: The graph of $f(x) = \cos(x)$ near $x = 0$

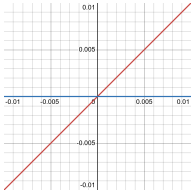



Figure: The graph of $f(x) = x$ near $x = 0$

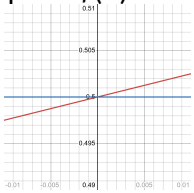



Figure: The graph of a logistic function near $x = 0$

# What to do?

So if constant functions $y = f(a) \approx f(x)$ don't approximate well enough what's the next most complicated type of function we could approximate $f(x)$ by?
*A linear function!*

# Linear Approximation

We will introduce the (extremely *extremely* important) concept of the derivative via *linear approximations*.

**Goal**

Find the linear function $\mathscr{L}(x) = mx + b$ that "*best*" approximates the function $f(x)$ when $x \approx a$ for some fixed value $a$.

Why? Oftentimes for scientific/engineering purposes, the actual functions that model complex phenomena (like pendulum forces, population growth rates, reaction rates, etc) are so wildly complicated that we cannot effectively study their true form.
However, taking their linear approximation (since linear functions are exactly specified by a finite amount of data [the slope and the intercept, e.g.]) often both simplify calculations vastly *AND* approximate real world measurements well enough.

# Best Linear Approximation I

(Aside: Because we want to look at behavior when $x \approx a$, I'll switch to *point-slope form* $\mathscr{L}(x) = m(x-a)+b$)

Idea 1: We'll say that $\mathscr{L}(x) = m(x-a)+b$ is the best linear approximation of $f(x)$ near $x=a$ if $\lim_{x\to a} \mathscr{L}(x) = \lim_{x\to a} f(x)$. Pros/Cons? This at least forces $b = \lim_{x\to a} f(x)$.

Idea 2: ... if the error term $|f(x) - \mathscr{L}(x)| \to 0$ as $x \to a$. But this is just the last idea dressed in a different hat...

We can't *just* make conditions on limits of errors since the information about the linear term $m(x-a)$ "*dies*" as $x \to 0$.

# Best Linear Approximation II

Solution: The error $|f(x) - \mathscr{L}(x)|$ should "die" to 0 faster than $|x - a|$ "dies" to 0.

**Best Linear Approximation**

A linear function $\mathscr{L}(x) = m(x - a) + b$ is the *best linear approximation* to a function $f(x)$ at $x = a$ if:

$$\lim_{x \to a} \frac{|f(x) - \mathscr{L}(x)|}{|x - a|} = 0 \tag{42}$$

That's great and all - but how do we *find* the best linear approximation?

# Best Linear Approximation III

## Determining the Best Linear Approximation

Given a function $f(x)$, if a best linear approximation $\mathscr{L}(x) = m(x-a)+b$ exists to $f(x)$ at $x=a$ then the following must be true:

1. $b = \lim_{x\to a} f(x)$
2. $f(x)$ is continuous at $x=a$, so
$$b = \lim_{x\to a} f(x) = f(a) \tag{43}$$
3. We have the following expression for the slope $m$:
$$m = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = IV_{x=a}(f(x)) \tag{44}$$

Because this quantity has shown up sooooo many times, we give it a special name:

**Derivative of a function at a point**

The derivative of a function $f(x)$ at a point $x = a$, denoted $f'(a)$ is the common value of the following four (equal) quantities:

1. $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{45}$$
2. The slope of the best linear approximation to $f(x)$ at $x = a$.
3. The slope of the *tangent* line to the graph of $y = f(x)$ at $x = a$.
4. The instantaneous velocity of a particle with position described by $f(x)$ at time $x = a$.

# Computing The Derivative

We often use Equation 45 to compute the derivative, which is why we spent the first two weeks going over limits in such depth!
E.g. let's compute the derivative of the function $f(x) = x^2$ at $x = 2$!

$$
\begin{aligned}
f'(2) &= \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2} && (46)\\
&= \lim_{x \to 2} \frac{x^2 - 2^2}{x - 2} && (47)\\
&= \lim_{x \to 2} \frac{(x-2)(x+2)}{x - 2} && (48)\\
&= \lim_{x \to 2} x + 2 = 2 + 2 = 4 && (49)
\end{aligned}
$$

$$
\begin{aligned}
f'(2) &= \lim_{h \to 0} \frac{f(2+h) - f(2)}{h} && (50)\\
&= \lim_{h \to 0} \frac{(2+h)^2 - 2^2}{h} && (51)\\
&= \lim_{h \to 0} \frac{4 + 4h + h^2 - 4}{h} && (52)\\
&= \lim_{h \to 0} \frac{4h + h^2}{h} && (53)\\
&= \lim_{h \to 0} 4 + h = 4 + 0 = 4 && (54)
\end{aligned}
$$

# Interpreting the Derivative

Does this actually give the slope of the best linear approximation/tangent line?

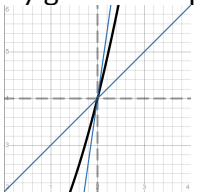


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 1, 7$

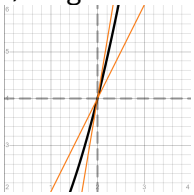


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 2, 6$

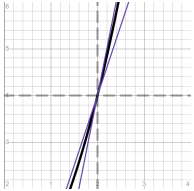


Figure: Linear approximations of $f(x) = x^2$ around $x = 2$ with slopes $m = 3, 5$

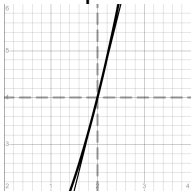


Figure: The best linear approximation of $f(x) = x^2$ around $x = 2$ with slope $m = f'(2) = 4$.

# Computing the Derivative II

Let's try another one! Let's compute $f'(4)$ when $f(x) = \sqrt{x}$.

$$f'(4) = \lim_{x \to 4} \frac{f(x) - f(4)}{x - 4} \tag{55}$$

$$= \lim_{x \to 4} \frac{\sqrt{x} - \sqrt{4}}{x - 4} \tag{56}$$

$$= \lim_{x \to 4} \frac{\sqrt{x} + \sqrt{4}}{\sqrt{x} + \sqrt{4}} \frac{\sqrt{x} - \sqrt{4}}{x - 4} \tag{57}$$

$$= \lim_{x \to 4} \frac{(\sqrt{x})^2 - (\sqrt{4})^2}{(x - 4)(\sqrt{x} + \sqrt{4})} \tag{58}$$

$$= \lim_{x \to 4} \frac{x - 4}{(x - 4)(\sqrt{x} + \sqrt{4})} \tag{59}$$

$$= \lim_{x \to 4} \frac{1}{\sqrt{x} + \sqrt{4}} = \frac{1}{\sqrt{4} + \sqrt{4}} = \frac{1}{4} \tag{60}$$

$$f'(4) = \lim_{h \to 0} \frac{f(4 + h) - f(4)}{h} \tag{61}$$

$$= \lim_{h \to 0} \frac{\sqrt{4 + h} - \sqrt{4}}{h} \tag{62}$$

$$= \lim_{h \to 0} \frac{(4 + h) - 4}{(\sqrt{4 + h} + \sqrt{4})h} \tag{63}$$

$$= \lim_{h \to 0} \frac{h}{(\sqrt{4 + h} + \sqrt{4})h} \tag{64}$$

$$= \lim_{h \to 0} \frac{1}{\sqrt{4 + h} + \sqrt{4}} = \frac{1}{4} \tag{65}$$

# Computing the Derivative III

For one last one, let's compute $f'(0)$ when $f(x) = |x|$.

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \tag{66}$$

$$= \lim_{x \to 0} \frac{|x|}{x} \tag{67}$$

since $|x|$ is defined piecewise, let's look at one-sided limits: (68)

$$\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} \tag{69}$$

$$= \lim_{x \to 0^+} 1 = 1 \tag{70}$$

$$\lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} \tag{71}$$

$$= \lim_{x \to 0^-} -1 = -1 \tag{72}$$

Since $\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} \neq \lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0}$ this means that $f'(0)$ *does not exist* and there is ***NO*** best linear approximation to $|x|$ around $x = 0$.

# Differentiable Functions I

## Differentiable Function

A function $f(x)$ is *differentiable* at $x = a$ if any of the following equivalent conditions hold:

1. The limit
$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{73}$$
exists.
2. There exists a best linear approximation $\mathscr{L}(x) = f(a) + f'(a)(x - a)$ to $f(x)$ around $x = a$.
3. There is a unique tangent line to the graph of $y = f(x)$ at $x = a$.

# Differentiable Functions II

The two classes of functions we've defined so far (continuous & differentiable) are related in the following manner:

## Continuity and Differentiability

For a function $f(x)$ and a point $x = a$ we have the following (equivalent) relationships:

1. If $f(x)$ is differentiable at $x = a$ then it *must* be the case that $f(x)$ is also continuous at $x = a$.
2. If $f(x)$ is *not* continuous at $x = a$ then it *can never* also be the case that $f(x)$ is also differentiable at $x = a$.

# Intuition for Continuity/Differentiability

The following intuition may be useful:

## Intuition for Continuity and Differentiability

We can intuitively recognize continuity/differentiability of a function $f(x)$ at a point $x = a$ from its graph as:

1. continuity: if you can draw the graph of $f(x)$ near $x = a$ *without* lifting up your pencil (no jump discontinuities) or without rapidly oscillating it (no $\sin(\frac{1}{x})$)
2. differentiability: if the graph of $f(x)$ "locally looks like" the graph of a linear function near $x = a$ i.e. the function is both continuous and does not make any sharp changes near $x = a$.

# Terminological Potpourri I

There's a lot of terminology so here's some scattered parts that didn't fit nicely.

## Tangent Line

Given a function $f(x)$ and a point $x = a$, the *tangent line* to $f(x)$ at $x = a$ is the line that:

1. intersects the graph of $y = f(x)$ at $(a, f(a))$
2. but does not *cross* the graph at that point (e.g. it just "glances" off).

The tangent line *is* the graph of the best linear approximation to $f(x)$ at $x = a$, as such the equation of the tangent line, for $f(x)$ differentiable at $x = a$, is:

$$y = f(a) + f'(a)(x - a). \tag{74}$$

# Terminological Potpourri II

**Units of the Derivative**

If our function represents the relationship between an input quantity $x$ measured in units of $U_{in}$ and an output quantity $f(x)$ measured in units of $U_{out}$ the derivative $f'(a)$ (if it exists) is measured in units of $\frac{U_{out}}{U_{in}}$.

E.g. if $s(t)$ is the distance in miles that a car travels, where $t$ is the time since starting measured in hours, $s'(t) = v(t)$ is the velocity and is measured in units of miles per hour.

# Worksheet Time!

Let's get into groups and work on the worksheet!

# End of Class!

That’s all folks!

# Lecture 6 - The Derivative as a Function

# Lecture 6 - The Derivative as a Function

Garrett Credi

Summer 2026

# Quiz Time!

Please get into organized rows so that you have a “bubble” of space around you away from other people.
Reminder:

1. Do not start the quiz until I announce you can start.
2. Do not read off other students’ work to cheat
3. Place all class material (notes, worksheets, etc) and devices (phones, tablets, headphones, etc) away
4. Don’t cheat and or be shady... obviously...

# Last Time

Last lecture we:

1. Introduced the idea of linearly approximating functions as an improvement to the constant-value approximation $f(x) \approx f(a)$ for $x \approx a$ of continuous functions.
2. Defined the *derivative* of a function $f(x)$ at a point $x = a$ as

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{75}$$

3. Saw that this quantity was also:
   (1) The slope of the best linear approximation

$$\mathscr{L}_{x=a}(x) = f(a) + f'(a)(x - a) \tag{76}$$

   to $f(x)$ at $x = a$.
   (2) The slope of the tangent line to the graph of $y = f(x)$ at $(a, f(a))$.
   (3) The instantaneous velocity of a particle whose motion is described by $f(x)$ at $x = a$.

# Last Time Cont.

We also were able to compute some derivatives using strategies from our limit section of the course e.g. for $f(x) = x^2$:

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \quad (77)$$

$$= \lim_{x \to 0} \frac{x^2}{x} \quad (78)$$

$$= \lim_{x \to 0} x = 0 \quad (79)$$

$$f'(1) = \lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} \quad (80)$$

$$= \lim_{x \to 1} \frac{x^2 - 1}{x - 1} \quad (81)$$

$$= \lim_{x \to 1} x + 1 = 1 + 1 = 2 \quad (82)$$

$$f'(2) = \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2} \quad (83)$$

$$= \lim_{x \to 2} \frac{x^2 - 4}{x - 2} \quad (84)$$

(85)

Wait... these all seem pretty similar!

# The Derivative Function I

What kind of mathematical object do we use when we want to compactly describe the process of taking some input value $a$ and computing an output quantity $f'(a)$?
A function!
Let's define *a derivative function* to see if we can simplify and speed up computations:

## The Derivative Function

For a function $f(x)$ we define the *derivative* (function), denoted either as:

$$f'(x) \text{ or } \frac{df}{dx} \text{ or } \frac{d}{dx}f(x),$$

to be the function so that:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{u \to x} \frac{f(u) - f(x)}{u - x} \tag{86}$$

Is this different from what we were doing before?

# The Derivative Function II

Yes! We can now compute *every* value of a derivative at once! E.g. for $f(x) = x^2$ again

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{87}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \tag{88}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \tag{89}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h} \tag{90}$$

$$= \lim_{h \to 0} 2x + h = 2x + 0 = 2x \tag{91}$$

So we immediately get that:

1. $f'(0) = 2 \cdot 0 = 0$
2. $f'(1) = 2 \cdot 1 = 2$
3. $f'(2) = 2 \cdot 2 = 4$

# The Derivative Function III

## Big Idea of Making it a Function

Here're two very important reasons why we might want to treat the derivative as a function and not just compute value by value:

1. Speed. Once we do the (slightly length) work of computing the general functional form of the derivative all future derivatives at a point can be computed *so quickly!* just by evaluation!
2. Information Extraction. Without the equation of the function, it's *impossible* to analytically study interesting properties of the derivative like:
   (1) For which values is $f'(a)$ positive? (when is $f(x)$ increasing?)
   (2) For which values is $f'(a)$ zero? (important for maximizing/minimizing which we'll do in 3/4 weeks)
   (3) For which values is $f'(a)$ undefined? (when can we not linearly approximate?)

   but given an equation of a function it's fairly easy to answer all of those!

# The Derivative Function IV

For example: at which $x$ values is the slope of the line tangent to $f(x) = x^3 - 2x + 1$ equal to zero?

Step 1: Translate to derivative question: when is $f'(x) = 0$?

Step 2: Compute derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{92}$$

$$= \lim_{h \to 0} \frac{(x+h)^3 - 2(x+h) + 1 - (x^3 - 2x + 1)}{h} \tag{93}$$

$$= \lim_{h \to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - 2x - 2h + 1 - x^3 + 2x - 1}{h} \tag{94}$$

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - 2h}{h} \tag{95}$$

$$= \lim_{h \to 0} 3x^2 + 3xh + h^2 - 2 \tag{96}$$

$$= 3x^2 + 3x \cdot 0 + 0^2 - 2 = 3x^2 - 2 \tag{97}$$

# The Derivative Function V

Last slide: $f'(x) = 3x^2 - 2$.
Step 3: Answer derivative question: when is $f'(x) = 3x^2 - 2 = 0$?

$$3x^2 - 2 = 0 \tag{98}$$
$$3x^2 = 2 \tag{99}$$
$$x^2 = \frac{2}{3} \tag{100}$$
$$x = \pm\sqrt{\frac{2}{3}} \tag{101}$$

Step 4: Translate derivative question answer to original question answer:

When $x$ is $\pm\sqrt{\frac{2}{3}}$, the slope of the line tangent to $f(x) = x^3 - 2x + 1$ is equal to zero.

# The Derivative Function VI

This is the general strategy that you can use to solve *sooo* many questions about the behavior of functions!
We'll see that in the worksheet for today :)

# The Derivative Function VII

Some conceptual questions:

1. For some random function $f(x)$ whose *domain* is all real numbers $\mathbb{R}$ must it also be the case that the domain of $f'(x)$ is also $\mathbb{R}$?
2. How do the following conditions relate:

(1) $f(x)$ is defined at $x = a$
(2) $\lim_{x \to a} f(x)$ exists
(3) $f(a) = \lim_{x \to a} f(x)$
(4) The domain of $f(x)$ is all real numbers $\mathbb{R}$
(5) $\lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}$ exists
(6) $f'(a)$ exists
(7) $f'(x)$ is defined at $x = a$
(8) $f(x)$ is differentiable at $x = a$
(9) $f'(x)$ is continuous at $x = a$
(10) $f'(x)$ is differentiable at $x = a$
(11) The domain of $f'(x)$ is all real numbers $\mathbb{R}$

3. Which of the above conditions relied on $f'(x)$ *being a function* rather than being a limit defined at a point $x = a$?

# Worksheet Time!

Let's get into groups and work on the worksheet!

# End of Class!

That’s all folks :)

# Lecture 7 - Interpreting, Estimating, and Applying the 1st Derivative and the 2nd Derivative

# Lecture 7 - Interpreting, Estimating, and Applying the 1st Derivative and the 2nd Derivative

Garrett Credi

Summer 2026

# Announcements

Next week, during class time we will have our first exam! It will cover material from sections 1.1-1.7 and 3.2.3 of Active Calculus. I.e. all the lectures up to and including this Thursday's lecture.

Practice exams and the review sheet are present in the Canvas.

How to study?

- Practice, practice, and more practice! The homework problems are your friends - use them to get practice solving the relevant problems.
- Lecture Slides for review. I'll upload a compiled version of all the slides with a searchable index at the bottom so you can quickly go to the slides defining relevant terms.
- Use the practice exams as a *cumulative metric* not as an additional source of problems. Take them after you complete a round of review, actually take them in an exam-like environment (no notes, 50 minutes, uncomfy chair, etc), and use your performance on them to guide your next round of review.
- Start early! The later you start, the less time you'll have to ask me questions in office hours/via email!
- Review session - I'll hold an Exam 1 review session next Monday (from 6-8?) where you can come and ask any and all questions you have and I'll answer em!

# Quiz Time!

Please get into neat rows, and spread out. Only then will I hand out the quiz. As per usual, no cheating (hopefully should go without saying).

# Derivative ReCap

Last week we studied the *derivative* of a function $f(x)$:

- At a point $x = a$, we defined it as

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{102}$$

- We saw that it was also equal to:
  1. The slope of the best linear approximation to $f(x)$ at $x = a$
  2. The instantaneous velocity of a particle whose position was given by $f(x)$ at $t = a$
  3. The slope of the line tangent to the graph of $y = f(x)$ at $(a, f(a))$.
- We also saw that we could construct a *derivative function*:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{103}$$

  and determine some information about $f(x)$ from $f'(x)$.

# Increasing-ness, Decreasing-ness, and Stationary-ness I

We can use the derivative to determine whether a function is increasing/decreasing/stationary:

**Increasing Function**

A function $f(x)$ is increasing around a point $x = a$ if it is increasing in *some* small neighborhood around $x = a$.

Decreasingness is similarly defined.

**Increasingness via the Derivative**

For a function $f(x)$ which is differentiable at $x = a$, increasingness at $a$ is equivalent to $f'(a) > 0$. Similarly, decreasingness is equivalent to $f'(a) < 0$.

# Increasingness II

Why is that true?
Idea: Because $f(x)$ is differentiable at $x = a$, it's behavior is well approximated by the best linear approximation $\mathscr{L}(x) = f(a) + f'(a)(x - a)$.
When is a linear function increasing/decreasing? When the slope (which is $f'(a)$) is positive/negative!
In terms of instantaneous velocity: positive velocity means increasing position, resp. negative velocity means decreasing position.

# Graph Sketching I

We can use this idea to sketch graphs of derivative functions $f'(x)$ given graphs of $f(x)$:

And to sketch graphs of function $f(x)$ given graphs of their derivatives $f'(x)$:

# Graph Sketching II

Notice how the increasingness/decreasingness of the *derivative* (e.g. the derivative of the derivative) informed us about the shape of the original function $f(x)$ - we'll see later in lecture this is due to the properties of the *second* derivative $f''(x)$.

# Estimation I

Oftentime, in the real world, we won't have a literal function of which to take the derivative we'll instead have some (possibly large) collection of sampled datapoints.
How can we get information about the derivative of the function modeling the measured quantity from these sparse datapoints?
The derivative is a limit so:

$$\text{Estimating the derivative} \iff \text{Estimating a limit!} \tag{104}$$

# Estimation II

If we're trying to estimate:

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{105}$$

We can compute the difference quotient quantities for values of $x \approx a$ or $h \approx 0$.
However, practically, there is a better choice of points about which to approximate:

## Central Difference Approximation

For a collection of sampled input and output pairs $(x_i, y_i)$, estimating the derivative of $y$ with respect to $x$ about $x = a$, $\frac{dy}{dx}|_{x=a}$ is best achieved by taking a pair of points $(x_0, y_0)$ and $(x_1, y_1)$ so that:

1. $x_0 \approx a$ and $x_1 \approx a$ (approximation requirement)
2. $x_0$ and $x_1$ are "symmetric" about $x = a$, i.e. one of them is close and slightly less than $a$ while the other is close and slightly greater than $a$.

# Central Difference Approximation Example

E.g.

# Graph Sketching Redux

Recall that when we sketched graphs of $f(x)$ given the graph of $f'(x)$

in addition to the positivity/negativity of $f'(x)$ giving us information about the increasingness/decreasingness of $f(x)$ the increasingness/decreasingness of $f'(x)$ itself also gave us information about the curvature of $f(x)$!

# The Second Derivative

The quantity that controls the increasingness/decreasingness of the *first* derivative $g(x) := f'(x)$ is *it's* derivative: $g'(x) = f''(x)$.

## Second Derivative

For a function $f(x)$ that is differentiable at $x = a$ and so that $f'(x)$ is *continuous* at $x = a$ (which we say $f(x)$ is continuously differentiable at $x = a$) , the second derivative of $f(x)$ at $x = a$ denoted:

$$f''(x) \quad f^{(2)}(x) \quad \frac{d^2 f}{dx^2} \quad \frac{d^2}{dx^2} f(x) \tag{106}$$

is defined to be the derivative of the function $f'(x)$ at $x = a$.
The second derivative function is defined similarly.

# Quadratic Approximation

Just as the first derivative controls information about the best *linear approximation*:

$$f(x) \approx f(a) + f'(a)(x - a) \tag{107}$$

The second derivative controls information about the best *quadratic approximation*:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 \tag{108}$$

Similarly, just as the first derivative is equivalent to instantaneous *velocity* , the second derivative is equivalent to instantaneous *acceleration*!

# Concavity I

| | |
|---|---|
| $f(x) > 0 / f(x) < 0$ | positivity/negativity |
| $f'(x) > 0 / f'(x) < 0$ | increasingness/decreasingness |
| $f''(x) > 0 / f''(x) < 0$ | ? |

## Concavity

A function $f(x)$ is *concave up* at $x = a$ if, around $x = a$ the function is *greater* than it's best linear approximation:

$$f(x) > f(a) + f'(a)(x - a) \tag{109}$$

equivalently that it lies above it's tangent lines.
Similarly, it is *concave down* at $x = a$ if:

$$f(x) < f(a) + f'(a)(x - a) \tag{110}$$

i.e. that is lies below it's tangent lines.

# Concavity II

## Concavity in terms of the Second Derivative

A function $f(x)$ which is second differentiable at $x = a$ is concave up at $x = a$ if and only if $f''(a) > 0$.
Similarly it is concave down if and only if $f''(a) < 0$.

E.g.

# Classifying Shapes of Functions via the First and Second Derivatives

We generically have four types of shapes a function can take given information about it's first and second derivatives:

If $f'(x) > 0$ and $f''(x) > 0$:

If $f'(x) < 0$ and $f''(x) > 0$:

If $f'(x) > 0$ and $f''(x) < 0$:

If $f'(x) < 0$ and $f''(x) < 0$:

# Worksheet Time!

Let's get into groups and work on the worksheet!

# End of Class

That’s all folks!

# Lecture 8 - Continuity and Differentiability

# Lecture 8 - Continuity and Differentiability

Garrett Credi

Summer 2026

# Announcements

1. Exam 1 is next Tuesday (10/06/2026) in class.
2. Exam 1 Review Session is next Monday (10/05/2026) from 6:10-8:10 PM in Vincent Hall 1.
3. All quizzes will be graded by Friday Night. If you want me to look at your homeworks, please email me and I'll grade them specifically.
4. Practice Exam Solutions will be released after class today.
5. Compiled lecture slides also released after class today.

# Quiz Time

Quiz Time! Let's get into neat rows and spread out - only then will I pass out the quizzes. DO NOT START OR FLIP OVER THE QUIZ UNTIL I OFFICIALLY START THE QUIZ. If you have any questions about the rules for quizzes, refer to the slide in Lecture 2 or ask me directly. No cheating as per usual.

# Continuity Recap I

Recall our definitions of continuity for a function $f(x)$:

1. $f(x)$ is continuous at $x = a$ if:
2. $f(x)$ is continuous on $(a, b)$ if:
3. $f(x)$ is continuous *everywhere* if:
4. $f(x)$ is continuous if:

# Continuity Recap II

Also recall that: if $f(x)$ and $g(x)$ are continuous at $x = a$ and $h(x)$ is continuous at $f(a)$ then:

- $f(x) + g(x)$ is continuous at $x = a$
- $f(x) - g(x)$ is continuous at $x = a$
- $f(x) \cdot g(x)$ is continuous at $x = a$
- If $g(a) \neq 0$, then $\dfrac{f(x)}{g(x)}$ is continuous at $x = a$
- $h(f(x))$ is continuous at $x = a$.

# Continuity Recap III

Finally we determined a large family of continuous functions:

- Constant functions $f_c(x) = c$
- Identity function $f(x) = x$
- Polynomials
  $p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$
- Root functions $f(x) = \sqrt[n]{x}$
- Rational functions $\dfrac{p(x)}{q(x)}$
- Trigonometric functions
  $\sin(x), \cos(x), \tan(x), \arcsin(x)$ etc.
- Exponential functions $f(x) = c^x$
- Logarithmic functions $f(x) = \log_c(x)$

# Classifying Failures of Continuity: Jump Discontinuities

What does discontinuity look like?

## Jump Discontinuity

A function $f(x)$ has a *jump discontinuity* at $x = a$, if the value of $f(x)$ jumps as we pass over $x = a$. Explicitly:

1. The function limits towards a value from the left: $\lim_{x \to a^-} f(x)$ exists
2. The function limits towards a value from the right: $\lim_{x \to a^+} f(x)$ exists
3. Those values do not agree:

$$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x) \tag{111}$$

# Classifying Failures of Continuity: Jump Discontinuities Visually

Visually, jump discontinuities look like:

# Classifying Failures of Continuity: Removable Discontinuities

Best kind of discontinuity.

**Removable Discontinuity**

A function $f(x)$ has a *removable discontinuity* at $x = a$ if:

1. $\lim_{x \to a} f(x)$ exists
2. $f(a)$ either does not exist (we specifically call this a *hole discontinuity* or just a *hole*) or is not equal to $\lim_{x \to a} f(x)$.

Can redefine $f(x)$ *at a single point* to make it continuous at $x = a$.

# Removable Discontinuities Visually

Visually,

# Classifying Failures of Continuity: Infinite Discontinuities

**Infinite Discontinuity**

A function $f(x)$ has an *infinite discontinuity* if, equivalently:

- $f(x)$ has a vertical asymptote at $x = a$ or
- $\lim_{x \to a} |f(x)| = +\infty$

# Infinite Discontinuities Visually

Visually,

# Classifying Failures of Continuity: Essential Discontinuities

The *WORST* kinds of discontinuities.

## Essential Discontinuity

A function $f(x)$ has an *essential discontinuity* at $x = a$ if equivalently

- At least one of the one-sided limits do not exist (not even diverging to $\pm infty$) or
- As $x \to a$, $f(x)$ takes on every number in some interval $(v_{min}, v_{max})$ *infinitely many times*

# Essential Discontinuities Visually

Visually,

# What does Continuity buy us?

We already saw one important use of continuous functions:

## Direct Substitution

If $f(x)$ is continuous at $x = a$ then:

$$\lim_{x \to a} f(x) = f(a) \tag{112}$$

Continuous functions also have another very important property:

## Intermediate Value Theorem

If $f(x)$ is continuous on $(a, b)$ then (assuming $f(a) \leq f(b)$)
for any intermediate output $c$ so that $f(a) \leq c \leq f(b)$, there must an input $x_0$ so that $a \leq x_0 \leq b$ and $f(x_0) = c$.

Tagline: all intermediate output values must be hit by intermediate input values.

# IVT Example: Root finding I

Question: For the function

$$f(x) = x^3 - 3x^2 + 2x + 1, \tag{113}$$

which of the following intervals must contain a root of $f(x)$?

1. $[-2, -1]$
2. $[-1, 0]$
3. $[0, 1]$
4. $[1, 2]$

Approach 1: Solving. IDK how to factor general cubics!
Approach 2: IVT. We want to find an interval which contains a value $x_0$ so that $f(x_0) = 0$. Note: if $a < 0$ is negative and $b > 0$ is positive, then 0 is always inside $[a, b]$.

# IVT Example: Root Finding II

Goal: find which of the intervals

1. $[-2, -1]$
2. $[-1, 0]$
3. $[0, 1]$
4. $[1, 2]$

makes

$$f(x) = x^3 - 3x^2 + 2x + 1 \tag{114}$$

switch signs at the endpoints.

1. $f(-2) = -8 - 3 \cdot 4 - 2 \cdot 2 + 1 = -8 - 12 - 4 + 1 = -23$
2. $f(-1) = -1 - 3 - 2 + 1 = -5$
3. $f(0) = 1$ ROOT!
4. $f(1) = 1 - 3 + 2 + 1 = 1$
5. $f(2) = 8 - 12 + 4 + 1 = 1$

Only know that a root must be inside $[-1, 0]$.

# Differentiability Recap

Again, recall that:

1. $f(x)$ is differentiable at $x = a$:
2. $f(x)$ is differentiable on $(a, b)$:
3. $f(x)$ is differentiable everywhere:
4. $f(x)$ is differentiable

# Classifying Failures of Differentiability: Discontinuities

We can already classify some behavior making functions differentiable.

Recall: if $f(x)$ is not *continuous* at $x = a$ then it cannot be *differentiable* at $x = a$.

# Classifying Failures of Differentiability: Corners

We can classify the rest of the behavior by examining the *continuity* of the derivative function $f'(x)$:

**Corner Singularity**

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *corner singularity* or *corner* at $x = a$ if the derivative function $f'(x)$ has a *jump discontinuity* at $x = a$.

Visually,

# Classifying Failures of Differentiability: Cusps

Similarly,

**Cuspidal Singularity**

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *cuspidal singularity* or *cusp* at $x = a$ if the derivative function $f'(x)$ has an *infinite discontinuity* at $x = a$ *AND* does not diverge to $\pm\infty$. (e.g. the one sided limits separately go to $+\infty$ and $-\infty$).

Visually,

# Classifying Failures of Differentiability: Vertical Tangents

## Vertical Tangent Singularity

For a function $f(x)$ continuous at $x = a$, we say $f(x)$ has a *vertical tangent singularity* or a *vertical tangent* at $x = a$ if:

$$\lim_{x \to a} |f'(x)| = +\infty. \tag{115}$$

Visually,

# Classifying Failures of Differentiability: Endpoint Singularity

Kind of a non-problem singularity:

**Endpoint Singularity**

For a function $f(x)$ defined on a *closed* interval $[a, b]$, we say that $f(x)$ has an *endpoint singularity* at the endpoints $x = a$ and $x = b$ just because two sided limits do not make sense at those points.

Visually,

# Continuous Differentiability I

What about the last kind of discontinuity: removable discontinuities?
The last thing that could go wrong would be if $f'(x)$ had a hole discontinuity at $x = a$ - not often discussed as a class of singularity.
Instead, we care when everything goes *right*:

## Continuous Differentiability

A function $f(x)$ is *continuously differentiable* at $x = a$ if $f'(x)$ is continuous at $x = a$.
Similarly, continuously differentiable on $(a, b)$, , everywhere, etc.
A function $f(x)$ is *continuously differentiable* or *of class* $C^1$ if is is continuously differentiable at every point in its domain.

Intuituion: Functions whose graphs are continuous, and corner/cusp/vertical tangent free.

# What does Continuous Differentiability buy us? I

Since $f(x)$ being continuously differentiable implies $f'(x)$ is continuous, we get the following by applying IVT to $f'(x)$:

**Rolle's Theorem**

For $f(x)$ continuously differentiable, and $x_0$ and $x_1$ two roots of $f(x)$ so that $x_0 \leq x_1$, there always exists an $a$ such that $x_0 \leq a \leq x_1$ and $f'(a) = 0$.

# What does Continuous Differentiability buy us? II

This also gives us a *VERY* important theorem:

## Mean Value Theorem

For $f(x)$ continuously differentiable, and $(x_0, f(x_0)), (x_1, f(x_1))$ so that $x_0 < x_1$ there is always an $a$ so that $x_0 \leq a \leq x_1$ and, equivalently:

- The tangent line $y = f'(a)(x - a) + f(a)$ is parallel to the secant line $y = \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + f(x_0)$ or
- $$\frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(a). \tag{116}$$

# Uses of the Mean Value Theorem

Why care about MVT?

1. If $f'(x)$ is the zero function $f'(x) = 0$, then $f(x)$ must be constant!

2. Extremely useful for doing derivations involving derivatives: e.g. determining quadratic approximations!

3. Lots of the really useful tools in multivariable calculus need to use MVT to even work.

“Fun” application:

# Worksheet Time

Let's get into groups and work on the worksheet!

# End of Class!

That’s all folks!

# Lecture 9 - Linearization and Derivative Rules

# Lecture 9 - Linearization and Derivative Rules

Garrett Credi

Summer 2026

# Post Exam Recap

1. Solutions to Exam 1 are up on Canvas, I'm happy to answer any questions about it if people want
2. Exams should be graded by tonight or tomorrow night, with curve details to follow after I release the scores
3. One important note so far: follow the instructions in the questions! Many people did not include reasoning for parts b and c of question 2; this will decrease your scores!

# Quiz Time!

Let’s get into neat rows and spread out for the quiz!
Just kidding :) no quiz today

# Best Linear Approximation

Recall that the *derivative* of a function $f(x)$ at a point $x = a$ was equal to the slope of the *best linear approximation* to $f(x)$ at $x = a$.

$$x \approx a \quad \Longrightarrow \quad f(x) \approx f(a) + f'(a)(x - a). \tag{117}$$

**Linearization**

The linearization of a differentiable function $f(x)$ at a point $x = a$ is the best linear approximation to $f(x)$ at $x = a$ and is equal to

$$\mathscr{L}_{x=a}(x) = f(a) + f'(a)(x - a) \tag{118}$$

# Using the Linearization to Approximate I

Using linearizations gives us *the strongest* ways approximate functions in a quick and accurate way.

**Example 1:** The *logistic population growth* model is a mathematical description of how the size of a population $P(t)$ grows over some time period $t$ with some growth factor $k$ and some carrying capacity $N$. It assumes that:

$$\frac{dP}{dt} = kP(N - P) \tag{119}$$

Suppose that we have a population $P(t)$, where $t$ is measured in days, which:

1. Starts with $1,000$ individuals So $P(0) = 1000$
2. Has a growth rate of $0.0001$ So $k = 0.0001$
3. Has a carrying capacity of $11,000$ individuals So $N = 11000$

What is our best estimate for the population size one year after we started measuring?

# Using the Linearization to Approximate II

$$\frac{dP}{dt} = kP(N-P), \quad P(0) = 1000, \quad k = 0.0001, \quad N = 11000, \quad P(1) \approx ? \tag{120}$$

Let's linearly approximate! The linearization of $P(t)$ around $t = 0$ is:

$$\mathscr{L}_{t=0}(t) = P(0) + P'(0)(t-0) \tag{121}$$

We know $P(0) = 1000$ what about $P'(0)$? We use the $\frac{dP}{dt}$ equation!

$$P'(0) = \frac{dP}{dt}\,|_{t=0} = k \cdot P(0) \cdot (N - P(0)) \tag{122}$$

$$= 0.0001 \cdot 1000 \cdot (11000 - 1000) = \frac{1}{10} \cdot 10000 \tag{123}$$

$$= 1000 \tag{124}$$

And so:

$$P(1) \approx P(0) + P'(0)(1-0) = 1000 + 1000 \cdot 1 = 2000 \tag{125}$$

# Using the Linearization to Approximate III

Is this an over-approximation ($\mathscr{L}_{t=0}(1) > P(1)$) or an under-approximation ($\mathscr{L}_{t=0}(1) < P(1)$)? What would we need to determine that?
Suppose we know that $P''(0) = 900$. Does that mean our linearization gave an over-approximation or an under-approximation?
Since $P''(0) > 0$, $P(t)$ is *concave-up* at $t = 0$, which means our approximation is an *under* approximation!
(Concave up => graph is above its tangent line)

# Using the Linearization to Approximate IV

We can also approximate functions that are:

1. Easy to define but
2. Hard to evaluate

like $f(x) = \sqrt{x}$, around known values.

**Example 2:** Approximate $\sqrt{25.1}$.

Approximation should always have us think of linearization: what do we need to linearize?

We need:

1. An initial value $x = a$ around which to approximate where
2. We know $f(a)$ AND
3. We know $f'(a)$.

What is a value $a \approx 25.1$ where we know $\sqrt{a}$ and the derivative of the sqrt function at $x = a$?

# Using the Linearization to Approximate V

We know that:

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}} \tag{126}$$

so knowing $\frac{d}{dx}\,|_{x=a}\,\sqrt{x}$ is as hard as knowing $\sqrt{a}$. What should $a$ be?

We should set $a = 25$ since $25 \approx 25.1$ AND $\sqrt{25} = 5$ is easy!

Then we get:

$$\sqrt{25.1} \approx \sqrt{25} + \frac{1}{2 \cdot \sqrt{25}}(25.1 - 25) \tag{127}$$

$$= 5 + \frac{1}{2 \cdot 5} \cdot (0.1) \tag{128}$$

$$= 5 + \frac{1}{10} \cdot 0.1 \tag{129}$$

$$= 5 + 0.01 = 5.01 \tag{130}$$

# Using the Linearization to Approximate VI

Let's say we also knew that:

$$\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{-1}{4\sqrt{x^3}}. \tag{131}$$

Is our approximation then an over or under approximation?
Well:

$$\frac{d^2}{dx^2}\Big|_{x=5}\sqrt{x} = \left(-\frac{1}{4\sqrt{x^3}}\right)\Big|_{x=5} \tag{132}$$

$$= -\frac{1}{4\sqrt{125}} < 0 \tag{133}$$

So this is an *over* approximation since we are concave down at $x = 5$.

# Limit Definition is Hard

How can we get that $\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{-1}{4\sqrt{x^3}}$?

$$\frac{d}{dx}\frac{1}{2\sqrt{x}} = \lim_{h\to 0}\frac{\frac{1}{2\sqrt{x+h}} - \frac{1}{2\sqrt{x}}}{h} \tag{134}$$

$$= \frac{1}{2}\lim_{h\to 0}\frac{\frac{\sqrt{x+h}}{x+h} - \frac{\sqrt{x}}{x}}{h} \tag{135}$$

$$= \frac{1}{2}\lim_{h\to 0}\frac{\frac{x\sqrt{x+h}}{x(x+h)} - \frac{(x+h)\sqrt{x}}{x(x+h)}}{h} \tag{136}$$

$$= \frac{1}{2}\lim_{h\to 0}\frac{\sqrt{x^3+hx^2} - \sqrt{(x+h)^2\cdot x}}{hx(x+h)} \tag{137}$$

$$= \ldots \tag{138}$$

GROSS!!

# Derivative Rules Rule!

We'll see that a much easier way to compute this derivative involved using *derivative rules* - variants of the sum/product/quotient limit rules we discussed a couple weeks ago.
For a sneak peek:

$$\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{d}{dx}\frac{1}{2}\frac{1}{x^{\frac{1}{2}}} \tag{139}$$

$$= \frac{d}{dx}\frac{1}{2}x^{-\frac{1}{2}} \tag{140}$$

$$= \frac{1}{2}\frac{d}{dx}x^{-\frac{1}{2}} \tag{141}$$

$$= \frac{1}{2}\cdot -\frac{1}{2}\cdot x^{-\frac{1}{2}-1} \tag{142}$$

$$= \frac{-1}{4\sqrt{x^3}} \tag{143}$$

# Derivative Rules

These will provide a *synthetic* approach to computing derivatives, and are split up similar to our rules for limits: some rules tell us how our operation acts on "simple" things, the rest tell us how our operation acts on "combinations" of things.

Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

# Sum/Difference Rule

Perhaps the most straightforward derivative rule:

**Sum/Difference Rule**

The derivative of a sum/difference of functions is the sum/difference (resp.) of the derivatives of the functions:

$$\frac{d}{dx}(f(x) + g(x)) = \left(\frac{d}{dx}f(x)\right) + \left(\frac{d}{dx}g(x)\right). \tag{144}$$

Why? What is the slope of a sum of two linear functions?

$$(f(a) + f'(a)(x-a)) + (g(a)g'(a)(x-a)) = (f(a) + g(a)) + (f'(a) + g'(a))(x-a) \tag{145}$$

# Constant Multiple Rule

Similarly:

## Constant Multiple Rule

The derivative of a function multiplied by a constant is that constant multiplied by the derivative of the function:

$$\frac{d}{dx}(c \cdot f(x)) = c\frac{d}{dx}f(x) \tag{146}$$

Why? What is the slope of a constant multiple of a linear function?

$$c \cdot (f(a) + f'(a)(x - a)) = (c \cdot f(a)) + (c \cdot f'(a))(x - a) \tag{147}$$

## WARNING!

The factor must actually be a constant!

$$\frac{d}{dx}x^2 = 2x \neq x \cdot \frac{d}{dx}x = x \cdot 1 = x \tag{148}$$

# Example

We can use this to compute derivatives of functions like:

$$f(x) = 5x^3 + 3x^2 + 2x + 5 \tag{149}$$

Since we already know that

$$\frac{d}{dx}x^3 = 3x^2, \quad \frac{d}{dx}x^2 = 2x, \quad \frac{d}{dx}x = 1, \quad \frac{d}{dx}1 = 0 \tag{150}$$

we get:

$$\begin{aligned}
f'(x) &= \frac{d}{dx}(5x^3 + 3x^2 + 2x + 5) && (151)\\
&= \frac{d}{dx}5x^3 + \frac{d}{dx}3x^2 + \frac{d}{dx}2x + \frac{d}{dx}5 && (152)\\
&= \frac{d}{dx}(5 \cdot x^3) + \frac{d}{dx}(3 \cdot x^2) + \frac{d}{dx}(2 \cdot x) + \frac{d}{dx}(5 \cdot 1) && (153)\\
&= 5 \cdot \frac{d}{dx}x^3 + 3 \cdot \frac{d}{dx}x^2 + 2 \cdot \frac{d}{dx}x + 5 \cdot \frac{d}{dx}1 && (154)\\
&= 5 \cdot (3x^2) + 3 \cdot (2x) + 2 \cdot 1 + 5 \cdot 0 && (155)\\
&= 15x^2 + 6x + 2 && (156)
\end{aligned}$$

# Power Rule I

What about for more complicated polynomials e.g. what if the degree is $> 3$?
We know that:

$$\frac{d}{dx}x^3 = 3x^2 \tag{157}$$

$$\frac{d}{dx}x^2 = 2x \tag{158}$$

$$\frac{d}{dx}x^1 = 1 \tag{159}$$

$$\frac{d}{dx}x^0 = 0 \tag{160}$$

Can we see a pattern?
Guess: $\frac{d}{dx}x^n = n \cdot x^{n-1}$.

# Power Rule II

Why?

$$\frac{d}{dx}x^n = \lim_{h\to 0}\frac{(x+h)^n - x^n}{h} \tag{161}$$

$$= \lim_{h\to 0}\frac{_x^n + _x^{n-1}\cdot h + _x^{n-2}h^2 + \ldots + _x^0 h^n - x^n}{h} \tag{162}$$

$$= \lim_{h\to 0}(_x^{n-1}\cdot 1 + _x^{n-1}h + \ldots + _x^0 h^{n-1}) \tag{163}$$

$$= _x^{n-1} \tag{164}$$

What is the coefficient on $hx^{n-1}$ in $(x+h)^n$? It's $n$!

So we know that $\frac{d}{dx}x^n = nx^{n-1}$!

# Power Rule III

## Power Rule

$$\frac{d}{dx}x^n = nx^{n-1} \tag{165}$$

E.g.:

- $\frac{d}{dx}x^{100} =$
- $\frac{d}{dx}x^{67} =$
- $\frac{d}{dx}x$

# Power Rule IV

Note: We technically only showed the limit definition worked for *integer* powers ($x^1, x^2, x^3$) as opposed to fractional powers, i.e. roots ($x^{\frac{3}{2}}, \sqrt{x}, \sqrt[3]{x^2}$). However, with product rule or inverse rule it's easy enough to show that power rule also works for fractional, and even all real-valued power functions.

E.g. $\frac{d}{dx}x^{\pi} =$

**WARNING!!**

Power rule only applies to ***POWER FUNCTIONS*** i.e. $f(x) = x^c$ for some constant $c$.
It does NOT apply to *exponential functions* like $f(x) = c^x$! They may look similar, but their derivatives are VASTLY different!

# Exponential Rule I

What is the derivative of an exponential function?

$$\frac{d}{dx}c^x = \lim_{h\to 0}\frac{c^{x+h}-c^x}{h} \tag{166}$$

$$= \lim_{h\to 0} c^x \frac{c^h-1}{h} \tag{167}$$

$$= \left(\lim_{h\to 0}\frac{c^h-1}{h}\right)c^x \tag{168}$$

it is just some constant multiple of the original function!

What is the constant factor $\lim_{h\to 0}\frac{c^h-1}{h}$?

# Exponential Rule II

Define a constant $e$ so that $\lim_{h\to 0} \frac{e^h - 1}{h} = 1$ (i.e. $\frac{d}{dx} e^x = e^x$). Then:

$$\lim_{h\to 0} \frac{c^h - 1}{h} = \lim_{h\to 0} \frac{(e^{\ln(c)})^h - 1}{h} \tag{169}$$

$$= \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{h} \tag{170}$$

$$= \lim_{h\to 0} \frac{\ln(c)}{\ln(c)} \frac{e^{\ln(c)h} - 1}{h} \tag{171}$$

$$= \ln(c) \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{\ln(c)h} \tag{172}$$

$$= \ln(c) \lim_{z\to 0} \frac{e^z - 1}{z} = \ln(c) \cdot 1 = \ln(c) \tag{173}$$

# Exponential Rule III

This gives us:

**Exponential Rule**

The derivative of an exponential function, i.e. $c^x$ for a constant $c$ is:

$$\frac{d}{dx}c^x = \ln(c)c^x \tag{174}$$

Exponential functions growing rapidly => it's derivative grows very large very quickly.

Very different behavior to that for power rule $\frac{d}{dx}x^c = cx^{c-1}$.

# Exponential Rule IV

What is the $n$'th derivative of an exponential function $c^x$?

First derivative: $\ln(c)c^x$.

Second derivative: $\frac{d}{dx}(\ln(c)c^x) = \ln(c)\frac{d}{dx}c^x = \ln(c)^2c^x$

Third derivative: $\ln(c)^3c^x$.

...

$n$'th derivative: $\ln(c)^nc^x$.

# Worksheet Time!

Let's get into groups and work on the worksheet!

# End of Class

That’s all folks!

# Lecture 10 - Product and Quotient Rule, Derivatives of Trig Functions

# Lecture 10 - Product and Quotient Rule, Derivatives of Trig Functions

Garrett Credi

Summer 2026

# Exam 1 Recap

1. All grades for Exam 1 released, regrade requests open until next Monday.
2. After Monday, curve will be determined
3. Recap: Essential Discontinuity, Vertical Tangent Singularity, Vertical Asymptotes

Also, Anonymous Feedback Form.

# Quiz Time!

Let's get into neat rows, spread out, and slay this quiz :)

# Long Term Goals

Recall we're trying to build a *synthetic* approach to studying derivatives:
Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

# Product Rule I

We want to determine what $\frac{d}{dx}(f(x) \cdot g(x))$ is, hopefully in terms of simpler derivatives involving $f(x), g(x)$.
Is it just $f'(x)g'(x)$? NO NO NO NO NO!




Figure: My one wish willow is for you to not compute derivates of products like this!

# Product Rule II

Well, then what is it?
Approach 1: Linearization! We motivated the other rules by examining what the operation did to linear functions first - let's try that here!
Linearization of $f(x)$ at $x = x_0$: $f(x_0) + f'(x_0)(x - x_0)$,
Linearization of $g(x)$ at $x = x_0$: $g(x_0) + g'(x_0)(x - x_0)$,
Linearization of $f(x)g(x)$ at $x = x_0$:

$$f(x)g(x) \approx (f(x_0) + f'(x_0)(x - x_0)) \cdot (g(x_0) + g'(x_0)(x - x_0)) \tag{175}$$
$$= f(x_0)g(x_0) + f'(x_0)g(x_0)(x - x_0) + f(x_0)g'(x_0)(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \tag{176}$$
$$= f(x_0)g(x_0) + (f'(x_0)g(x_0) + f(x_0)g'(x_0))(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \tag{177}$$

So why is it NOT $f'(x)g'(x)$? Because that's the coefficient of something involving a *quadratic* term, not linear!
The derivative would be the slope on the *linear* term.

# Product Rule III

## The Product Rule

$$\frac{d}{dx}(f(x) \cdot g(x)) = f'(x)g(x) + f(x)g'(x) \tag{178}$$

# Product Rule Examples I

Example 1: What is the derivative of:

$$x^2e^x + \frac{2^x}{x}? \tag{179}$$

We compute:

$$\frac{d}{dx}(x^2e^x + \frac{2^x}{x}) = \frac{d}{dx}(x^2e^x) + \frac{d}{dx}(\frac{2^x}{x}) \tag{180}$$

$$= (\frac{d}{dx}x^2) \cdot (e^x) + (x^2) \cdot (\frac{d}{dx}e^x) + \frac{d}{dx}(2^x \cdot \frac{1}{x}) \tag{181}$$

$$= (2x) \cdot e^x + x^2 \cdot e^x + (\frac{d}{dx}2^x) \cdot (\frac{1}{x}) + (2^x) \cdot (\frac{d}{dx}\frac{1}{x}) \tag{182}$$

$$= 2xe^x + x^2e^x + \ln(2)2^x \cdot \frac{1}{x} + 2^x \cdot \frac{-1}{x^2} \tag{183}$$

# Product Rule Examples II

For a more applied example: Let's suppose you drive $m(t)$ miles per day, in a car whose fuel efficiency is given by $e(t)$ gallons/mile where the price of fuel is given by $p(t)$ dollars/gallon, $t$ is measured in days.
Suppose we've measured the relevant quantities at:

$$m(1) = 50mi, \quad e(1) = \frac{1}{25}\frac{gal}{mi}, \quad p(1) = 4.15\frac{\$}{gal} \tag{184}$$

$$m'(1) = 0\frac{mi}{day}, \quad e'(1) = \frac{1}{1000}\frac{\frac{gal}{mi}}{day}, \quad p'(1) = 0.01\frac{\frac{\$}{gal}}{day} \tag{185}$$

What is the rate of change of the cost to drive your car on day 1?

# Product Rule Examples III

Step 1: Cost Function. $C(t) = m(t) \cdot e(t) \cdot p(t)$.
Step 2: Derivative.

$$C'(t) = \frac{d}{dt}(m(t) \cdot e(t) \cdot p(t)) = \frac{d}{dt}((m(t)e(t)) \cdot p(t)) \tag{186}$$

$$= \frac{d}{dt}(m(t) \cdot e(t)) \cdot p(t) + (m(t) \cdot e(t)) \cdot p'(t) \tag{187}$$

$$= m'(t)e(t)p(t) + m(t)e'(t)p(t) + m(t)e(t)p'(t) \tag{188}$$

# Product Rule Examples IV

Step 3: Compute.

$$C'(1) = m'(1)e(1)p(1) + m(1)e'(1)p(1) + m(1)e(1)p'(1) \tag{189}$$

$$= 0 \cdot \frac{1}{25}\frac{gal}{mi} \cdot 4.15\frac{\$}{gal} \tag{190}$$

$$+ 50mi \cdot \frac{1}{1000}\frac{\frac{gal}{mi}}{day} \cdot 4.15\frac{\$}{gal} \tag{191}$$

$$+ 50mi \cdot \frac{1}{25}\frac{gal}{mi} \cdot 0.01\frac{\frac{\$}{gal}}{day} \tag{192}$$

$$= 0\frac{\$}{day} + 0.2075\frac{\$}{day} + 0.02\frac{\$}{day} \tag{193}$$

$$= 0.2275\frac{\$}{day} \tag{194}$$

# Quotient Rule I

Similarly, we'd want to determine $\frac{d}{dx}\frac{f(x)}{g(x)}$, again in terms of simpler derivatives involving $f(x)$, $g(x)$.

Is it just $\frac{f'(x)}{g'(x)}$? SUPER NO! NO! NO!




Figure: I'm amending my one wish willow to also include you not computing derivatives of quotients like this.

# Quotient Rule II

We can determine what it actually is by realizing that:

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} \tag{195}$$

and since we know how to linearize $f(x)$ and how to linearize a product, we need only understand how to linearize something like $\frac{1}{g(x)}$.

First, we note that for the derivative of $\frac{f(x)}{g(x)}$ at $x = x_0$ to exist, the function $\frac{f(x)}{g(x)}$ must be continuous at $x = x_0$, so $g(x_0) \neq 0$.

This allows us to approximate $\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)}$.

# Quotient Rule III

Since

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)} = \frac{1}{(g(x_0) - g'(x_0)x_0) + g'(x_0)x} \tag{196}$$

we can focus on understanding how to linearize functions like

$$\frac{1}{a + bx}. \tag{197}$$

We compute:

$$\frac{d}{dx}\frac{1}{a+bx} = \frac{d}{dx}\frac{1}{b}\frac{1}{x + \frac{a}{b}} = \frac{1}{b}\frac{d}{dx}\frac{1}{x + \frac{a}{b}} \tag{198}$$

$$= \frac{1}{b}\frac{-1}{(x + \frac{a}{b})^2} = \frac{1}{b}\frac{-1}{x^2 + \frac{2a}{b}x + \frac{a^2}{b^2}} \tag{199}$$

$$= \frac{-1}{bx^2 + 2ax + \frac{a^2}{b}} = \frac{-b}{b^2x^2 + 2abx + a^2} = \frac{-b}{(a+bx)^2} \tag{200}$$

# Quotient Rule IV

And so,

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0)} + \frac{-g'(x_0)}{g(x_0)^2}(x - x_0) \tag{201}$$

meaning that:

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} \tag{202}$$

$$= (f(x_0) + f'(x_0)(x - x_0)) \cdot (\frac{1}{g(x_0)} + \frac{-g'(x_0)}{g(x_0)^2}(x - x_0)) \tag{203}$$

$$= \frac{f(x_0)}{g(x_0)} + \frac{f'(x_0)}{g(x_0)}(x - x_0) + \frac{-f(x_0)g'(x_0)}{g(x_0)^2}(x - x_0) + O((x - x_0)^2) \tag{204}$$

$$= \frac{f(x_0)}{g(x_0)} + \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}(x - x_0) \tag{205}$$

# Quotient Rule V

Moral of the story: it's HARD to actually get the derivative of a quotient so in no universe would the formula be as easy as $\frac{f'(x)}{g'(x)}$. It is actually:

**Quotient Rule**

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{206}$$

# Quotient Rule Example I

E.g. computing:

$$\frac{d}{dx}\frac{2xe^x + 3x^2}{x^3 + 2x + 1} \tag{207}$$

$$\frac{d}{dx}\frac{2xe^x + 3x^2}{x^3 + 2x + 1} = \frac{(\frac{d}{dx}(2xe^x + 3x^2)) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (\frac{d}{dx}(x^3 + 2x + 1))}{(x^3 + 2x + 1)^2} \tag{208}$$

$$= \frac{(\frac{d}{dx}(2xe^x) + \frac{d}{dx}(3x^2)) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{209}$$

$$= \frac{(2e^x + 2xe^x + 6x) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{210}$$

# Derivative of $\sin(x)$ I

Recall from our discussion on the Squeeze Theorem:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \tag{211}$$

$$\lim_{x \to 0} \frac{\cos(x) - 1}{x} = 0 \tag{212}$$

What is this limit in terms of derivatives?

$$\frac{d}{dx} |_{x=0} \sin(x) = 1, \frac{d}{dx} |_{x=0} \cos(x) = 0 \tag{213}$$

# Derivative of $\sin(x)$ II

What about the derivative function?

$$\frac{d}{dx}\sin(x) = \lim_{h\to 0}\frac{\sin(x+h) - sin(x)}{h} \tag{214}$$

$$= \lim_{h\to 0}\frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h} \tag{215}$$

$$= \lim_{h\to 0}\frac{\sin(x)(\cos(h) - 1) + \cos(x)\sin(h)}{h} \tag{216}$$

$$= \lim_{h\to 0}\left(\sin(x)\frac{\cos(h) - 1}{h} + \cos(x)\frac{\sin(h)}{h}\right) \tag{217}$$

$$= \sin(x)\lim_{h\to 0}\left(\frac{\cos(h) - 1}{h}\right) + \cos(x)\lim_{h\to 0}\left(\frac{\sin(h)}{h}\right) \tag{218}$$

$$= \sin(x)\cdot 0 + \cos(x)\cdot 1 = \cos(x) \tag{219}$$

# Derivative of $\sin(x)$ III

**Derivative of $\sin(x)$**

$$\frac{d}{dx}\sin(x) = \cos(x) \tag{220}$$

What about for $\cos(x)$?

# Derivative of $\cos(x)$ I

$$\frac{d}{dx}\cos(x) = \lim_{h\to 0}\frac{\cos(x+h)-\cos(x)}{h} \tag{221}$$

$$= \lim_{h\to 0}\frac{\cos(x)\cos(h)-\sin(x)\sin(h)-\cos(x)}{h} \tag{222}$$

$$= \cos(x)\lim_{h\to 0}\left(\frac{\cos(h)-1}{h}\right) - \sin(x)\lim_{h\to 0}\left(\frac{\sin(h)}{h}\right) = -\sin(x) \tag{223}$$

**Derivative of $\cos(x)$**

$$\frac{d}{dx}\cos(x) = -\sin(x) \tag{224}$$

# Derivative of $\tan(x)$

What about for $\tan(x)$? Do we need to do even more limit yuckiness?

No! Since $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$ we can just use quotient rule!

$$\frac{d}{dx}\tan(x) = \frac{d}{dx}\frac{\sin(x)}{\cos(x)} \tag{225}$$

$$= \frac{\left(\frac{d}{dx}\sin(x)\right)\cdot\cos(x) - \sin(x)\cdot\left(\frac{d}{dx}\cos(x)\right)}{\cos(x)^2} \tag{226}$$

$$= \frac{\cos(x)\cdot\cos(x) - \sin(x)\cdot(-\sin(x))}{\cos(x)^2} \tag{227}$$

$$= \frac{\cos(x)^2 + \sin(x)^2}{\cos(x)^2} \tag{228}$$

$$= \frac{1}{\cos(x)^2} = \sec(x)^2 \tag{229}$$

# Derivative of $\sin(2x)$

While we'll get an easier method to do this next lecture, we can also use trig rules to compute $\frac{d}{dx}\sin(2x)$:

$$\frac{d}{dx}\sin(2x) = \frac{d}{dx}(2\sin(x)\cos(x)) \tag{230}$$

$$= 2\frac{d}{dx}(\sin(x)\cos(x)) \tag{231}$$

$$= 2 \cdot \left(\left(\frac{d}{dx}\sin(x)\right) \cdot \cos(x) + \sin(x) \cdot \left(\frac{d}{dx}\cos(x)\right)\right) \tag{232}$$

$$= 2 \cdot (\cos(x) \cdot \cos(x) + \sin(x) \cdot (-\sin(x))) \tag{233}$$

$$= 2 \cdot (\cos(x)^2 - \sin(x)^2) \tag{234}$$

$$= 2 \cdot \cos(2x) \tag{235}$$

# Worksheet Time!

Let's get into groups (or if you really like working alone, please stay so I can be here to help :) ) and work on the worksheet!

# End of Class!

That’s all folks!

# Lecture 11 - The Chain Rule

# Lecture 11 - The Chain Rule

Garrett Credi

Summer 2026

# Quiz Time

Let's get into neat rows, spread out, and slay this quiz! :)

# The Road So Far

Recall we're trying to build a *synthetic* approach to studying derivatives:
Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

# What is the Chain Rule? I

We're spending today building up the chain rule, the tool that will help us compute derivatives of the form:

$$\frac{d}{dx}(g(f(x))) \tag{236}$$

But what kinds of functions will this actually help us compute derivative of?
Sum/Difference Rule and Product/Quotient rule have visible signifiers that help us determine when to use them:

But what do functions of the form $g(f(x))$ even look like?

# What is the Chain Rule? II

Recall our “basic” functions:

- $x^n$
- $c^x$
- $\sin(x), \cos(x)$, etc
- $|x|$
- $\log_c(x)$

When we *compose* functions (do the $g(f(x))$ thing), what does this actually do?
We make one function’s output another function’s input instead of just inputting $x$!

# What is the Chain Rule? III

So, before we even *use* chain rule, we should first come up with a strategy for:

1. Recognizing when functions are composite functions
2. Recognizing which function is the inner function (the $f(x)$ in the $g(f(x))$) and which function is the outer function (the $g(x)$ in the $g(f(x))$)

Strategy:

1. Break up the complicated function into the addition, subtraction, product, and quotient of simpler functions (i.e. do sum/difference and product/quotient rules first)
2. In each of the smaller pieces, identify where the variables actually are.
3. At each occurrence of a variable, find the biggest/longest expression that it lives in - this is the innermost function.
4. Treat that whole expression as a new variable, and repeat from step 1.

# Identifying Composites Example

For instance, let's take $h(t) = 3\sin^2(4t^2 + 1) \cdot \cos(4t^2 + 1) \cdot 8t$

# Deriving Chain Rule I

Now let's assume we've already identified some function of interest as a composite $g(f(x))$, we'd like to find an expression of the derivative $\frac{d}{dx}g(f(x))$ hopefully in terms of derivatives involving $f(x)$ and $g(x)$.
Strategy: Linearize once again! Let's linearize around some $x = x_0$.
We start by linearizing *only* $f(x)$, since our input to $g(x)$ is changed by $f$ and thus might end up close to a point different than $x = x_0$.
We know that $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$, so we end up wanting to further linearize:

$$g(f(x_0) + f'(x_0)(x - x_0))$$

# Deriving Chain Rule II

Note: when we linearize $f(x)$ near $x = x_0$, we mean that for values $x \approx x_0$ we'll know that $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$ i.e. our input value being close to something implies our output value is close to the related thing.
For

$$g(f(x_0) + f'(x_0)(x - x_0))$$

what value are we inputting into the function $g$?
When $x \approx x_0$, what is $f(x_0) + f'(x_0)(x - x_0)$ close to?
Since $f(x_0) + f'(x_0)(x - x_0) \approx f(x_0)$, we will linearize $g(x)$ around $f(x_0)$ instead!

# Deriving Chain Rule III

$$
\begin{aligned}
g(f(x)) &\approx g(f(x_0) + f'(x_0)(x - x_0)) \quad \text{linearize } g \text{ around } f(x_0) && (237)\\
&\approx g(f(x_0)) + g'(f(x_0))((f(x_0) + f'(x_0)(x - x_0)) - f(x_0)) && (238)\\
&= g(f(x_0)) + g'(f(x_0))(f'(x_0)(x - x_0) + f(x_0) - f(x_0)) && (239)\\
&= g(f(x_0)) + g'(f(x_0))f'(x_0)(x - x_0) && (240)
\end{aligned}
$$

# Deriving Chain Rule IV

The derivative is the slope of the best linear approximation $\implies$

**The Chain Rule**

$$\frac{d}{dx}g(f(x)) = g'(f(x))f'(x) \tag{241}$$

Or, written differently

**The Chain Rule (Leibniz's Version)**

$$\frac{d(g \circ f)}{dx}\Big|_{x=x_0} = \frac{dg}{dx}\Big|_{x=f(x_0)} \cdot \frac{df}{dx}\Big|_{x=x_0} \tag{242}$$

# Chain Rule Example I

Let's compute the derivative of that function from before:

$$h(t) = 3\sin^2(4t^2 + 1) \cdot \cos(4t + 1) \cdot 8t \tag{243}$$

# Chain Rule Example II

We can also derive the quotient rule in terms of the chain rule since

$$\frac{f(x)}{g(x)} = f(x) \cdot \frac{1}{g(x)} = f(x) \cdot (\frac{1}{x} \circ g)(x) \tag{244}$$

i.e. $\frac{1}{g(x)}$ is a composite function! You first do $g(x)$ and then do the function $\frac{1}{x}$.
Then:

$$\frac{d}{dx}\frac{f(x)}{g(x)} = \left(\frac{d}{dx}f(x)\right) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{d}{dx}\frac{1}{g(x)}\right) \tag{245}$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \left(\left(\frac{d}{du}\frac{1}{u}\right)|_{u=g(x)} \cdot \left(\frac{d}{dx}g(x)\right)\right) \tag{246}$$

$$= \frac{f'(x)}{g(x)} + f(x) \cdot \frac{-1}{g(x)^2} \cdot g'(x) = \frac{f'(x)g(x)}{g(x)^2)} - \frac{f(x)g'(x)}{g(x)^2} \tag{247}$$

$$= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \tag{248}$$

# Chain Rule Example III

We can also frame this more in terms of real world applications, too long for a slide but it's in the worksheet.

# Worksheet Time

Let's get into groups and work on the worksheet!

# End of Class

That’s all folks!

# Lecture 12 - Inverse and Implicit Differentiation

# Lecture 12 - Inverse and Implicit Differentiation

Garrett Credi

Summer 2026

# Updates

Curve information forthcoming - in contact with course coordinator about adjusting final grade distribution.
Regrade requests for Exam 1 closed.

# Quiz Time!

Let's get into neat rows, spread out, and slay this quiz!

# The Road So Far

Recall we're trying to build a *synthetic* approach to studying derivatives:
Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

# Explicit vs. Implicit Definitions I

Up until now we've been dealing with *explicit* definitions of various functions and have been using those *explicit* definitions to compute their derivatives.
An *explicit* definition of a function is one where we defined $f(x)$ by *explicitly* saying where a given input $x$ maps to. E.g.:

- $f(x) = x^2$
- $g(x) = \sin(x)$
- $h(x) = \sqrt{x^2 + 2x}$
- $j(x) = e^{\sin(x)}$
- $k(x) = (x + \sin(x))^{\frac{1}{2}}$
- etc

This is not the only way in which we can define functions!

# Explicit vs Implicit Definitions II

We could instead define relations *implicitly* - i.e. by writing out some equation that we want the function to satisfy, e.g.:

- $x^2 + y^2 = 1$
- $y^2 = x^3 + 2x + 1$
- $\arctan(\frac{y}{x}) = \frac{\pi}{4}$
- $\sin(x) + \cos(y) = 1$
- $x^2 - y^2 = 2$
- $x^3 + y^3 = 1$

Just as we can *graph* a function: i.e. we plot all the points $(x, f(x))$, we can plot some implicit equation $F(x, y) = 0$ by plotting all the points $(x, y)$ so that $F(x, y) = 0$.
When is this useful?

# Applications of Implicit Relations

Oftentimes, real world process don't have simple ways of relating one quantity to another quantity.
Consider: conservation of momentum/energy in physics:

$$\sum_{i=1}^{n} m_i v_i^{init} = \sum_{i=1}^{n} m_i v_i^{final} \tag{249}$$

$$\sum_{i=1}^{n} \frac{1}{2} m_i (v_i^{init})^2 = \sum_{i=1}^{n} \frac{1}{2} m_i (v_i^{final})^2 \tag{250}$$

The relation between initial and final velocities here are not a simple input-to-output relationship - they are equations that need to be enforced.
Also: predator-prey models, resource allocation, budgeting, etc.

# Inverse Functions as a Relation

To address our final desired synthetic rule (inverse function rule) we can also see that inverse functions are defined via an implicit rule:

## Inverse Function

For a function $f(x)$ with domain $D \subseteq \mathbb{R}$ and range $R \subseteq \mathbb{R}$, the inverse function $f^{-1}(x)$ is the function that:

- Has domain $R$ (domain of inverse is range of original)
- Has range $D$ (range of inverse is domain of original)
- Satisfies the two relations:
  1. $f(f^{-1}(x)) = x$
  2. $f^{-1}(f(x)) = x$

  i.e. $f(x)$ and $f^{-1}(x)$ "undo" eachother.

# Examples of Inverse Functions

For some examples:

1. If $f(x) = x^2$ with domain *the positive real numbers* and range *the positive real numbers* $f^{-1}(x) = \sqrt{x}$ is the positive square root.
2. If $f(x) = x^2$ with domain *the negative real numbers* and range *the positive real numbers* $f^{-1}(x) = -\sqrt{x}$ is the negative square root.
3. If $f(x) = \sin(x)$ with domain $[0, 2\pi)$ and range $[-1, 1]$ $f^{-1}(x) = \arcsin(x)$
4. If $f(x) = e^x$ with domain $\mathbb{R}$ and range $(0, \infty)$ $f^{-1}(x) = \ln(x)$.

# Inverse Differentiation I

Question: Using the relation

$$f(f^{-1}(x)) = x \tag{251}$$

how might we relate the derivative of $f(x)$ with the derivative of it's inverse function $f^{-1}(x)$?
Answer: Chain Rule!

$$f(f^{-1}(x)) = x \tag{252}$$

Differentiate both sides (253)

$$\frac{d}{dx} f(f^{-1}(x)) = \frac{d}{dx} x \tag{254}$$

$$\left(\frac{d}{dx} f^{-1}(x)\right) \cdot f'(f^{-1}(x)) = 1 \tag{255}$$

$$\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))} \tag{256}$$

# Inverse Differentiation II

## Inverse Differentiation

If $f(x)$ and $f^{-1}(x)$ are an inverse pair of functions then:

$$\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))} \tag{257}$$

# Inverse Differentiation III

Ex. $\frac{d}{dx}\ln(x)$.
$\ln(x)$ is inverse to $e^x \implies f(x) = e^x, f^{-1}(x) = \ln(x)$.

$$\frac{d}{dx}\ln(x) = \frac{1}{\left(\frac{d}{du}e^u\right)|_{u=\ln(x)}} = \frac{1}{(e^u)|_{u=\ln(x)}} \tag{258}$$

$$= \frac{1}{e^{\ln(x)}} = \frac{1}{x} \tag{259}$$

**Derivative of Natural Logarithm**

$$\frac{d}{dx}\ln(x) = \frac{1}{x} \tag{260}$$

# Inverse Differentiation IV

Ex. $\frac{d}{dx}\arcsin(x)$.

$$\frac{d}{dx}\arcsin(x) = \frac{1}{\left(\frac{d}{du}\sin(u)\right)|_{u=\arcsin(x)}} = \frac{1}{(\cos(u))|_{u=\arcsin(x)}} \tag{261}$$

$$= \cos(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}} \tag{262}$$

$$\frac{d}{dx}\arccos(x) = \frac{1}{-\sin(\arccos(x))} = -\frac{1}{\sqrt{1-x^2}} \tag{263}$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{\sec^2(\arctan(x))} = \frac{1}{1+x^2} \tag{264}$$

# Inverse Differentiation V

## Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\arcsin(x) = \frac{1}{\sqrt{1-x^2}} \tag{265}$$

$$\frac{d}{dx}\arccos(x) = -\frac{1}{\sqrt{1-x^2}} \tag{266}$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2} \tag{267}$$

# Implicit Differentiation I

What about for the other kinds of implicit relations?

Q: At a point $(x, y)$ on the circle of radius 1, what is the slope of the line tangent to the circle at that point?

Circle of radius 1 $\implies x^2 + y^2 = 1$.

$$x^2 + y^2 = 1 \tag{268}$$

Differentiate both sides w.r.t. $x$ (269)

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}1 \tag{270}$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0 \tag{271}$$

$$2x + \frac{d}{dx}(y(x)^2) = 0 \tag{272}$$

$$2x + 2y \cdot \frac{dy}{dx} = 0 \quad 2y\frac{dy}{dx} = -2x \quad \frac{dy}{dx} = -\frac{x}{y} \tag{273}$$

# Implicit Differentiation II

The core idea is that in an implicit relation $F(x, y) = 0$, we can *always* treat one of the variables as a function of the other (fancy reason: *implicit function theorem*) and thus can differentiate implicit relations just by using the chain rule!
We'll see more application oriented examples when we do related rates soon :)

# Worksheet Time

Let's get into groups and work on the worksheet!

# End of Class

That’s all folks!

# Lecture 13 - Related Rates

# Lecture 13 - Related Rates

Garrett Credi

Summer 2026

# Exam 2 Soon!

Exam 2 is coming up! It is two Tuesday's from now, 11/3.
It's covering linearizations + our derivative rules + related rates + L'Hôpital's Rule which we'll cover next week.

# Quiz Time

Let's get into neat rows, spread out, and slay this quiz!

# Implicit Differentiation Recap

Last lecture we covered *implicit differentiation*, the process of relating variables using some implicit relation and differentiating that relation to get a relation between their derivatives.
E.g.: slope of line tangent to $y^2 = x^3 - 10x + 1$ at $(0, 1)$.

Slope of tangent line $\implies$ interested in $\dfrac{dy}{dx}$ $\implies$ treat $y$ as a function of $x$.

$$y^2 = x^3 - 10x + 1 \tag{274}$$

$$\frac{d}{dx}y^2 = \frac{d}{dx}(x^3 - 10x + 1) \tag{275}$$

$$\frac{dy}{dx}\frac{d}{dy}y^2 = 3x^2 - 10 \tag{276}$$

$$2yy' = 3x^2 - 10 \tag{277}$$

$$2 \cdot 1 \cdot y' = 3 \cdot 0^2 - 10 \tag{278}$$

$$2y' = -10 \quad y' = -5 \tag{279}$$

# Towards Related Rates

Suppose instead that some particle was moving over time in the $2D$ plane and whose motion was constrained to that curve cut out by $y^2 = x^3 - 10x + 1$.

- Moving over time $\implies$ it's position can be described by functions of a time variable $t$
- In the $2D$ plane $\implies$ it has an $x$-position $x(t)$ and a $y$-position $y(t)$
- Motion constrained $\implies$ $y(t)^2 = x(t)^3 - 10x(t) + 1$.

In order to stay on that curve, what restrictions does that place on the particle's:

1. Position?
2. Velocity?
3. Acceleration?

# Relating Some Rates I

Let's see what our constraint on position forces upon our velocity!

$$y(t)^2 = x(t)^3 - 10x(t) + 1 \tag{280}$$

Differentiate w.r.t. $t$ to get velocity (281)

$$\frac{d}{dt}y(t)^2 = \frac{d}{dt}(x(t)^3 - 10x(t) + 1) \tag{282}$$

$$\frac{dy}{dt}\frac{d}{dy}y^2 = \frac{dx}{dt}\frac{d}{dx}(x^3 - 10x + 1) \tag{283}$$

$$2y\frac{dy}{dt} = (3x^2 - 10)\frac{dx}{dt} \tag{284}$$

# Relating Some Rates II

So if the particle is at $(0, 1)$ at some time $t = t_0$ and has a horizontal velocity of $3m/s$ which means $(x(t_0), y(t_0)) = (0, 1)$ and $\frac{dx}{dt}\,|_{t=t_0} = 3$ what does that force the vertical velocity to be?

$$2y\frac{dy}{dt} = (3x^2 - 10)\frac{dx}{dt} \tag{285}$$

$$2 \cdot 1\frac{dy}{dt} = (3 \cdot 0^2 - 10) \cdot 3 \tag{286}$$

$$2\frac{dy}{dt} = -30 \quad \frac{dy}{dt} = -15m/s \tag{287}$$

(Notice from before, we found the slope of the line tangent to the curve at $(1, 0)$ was $-5$ and $-15 = -5 \cdot 3$, so we're exactly enforcing that the velocity is parallel to the tangent line).

# Related Rates I

This process is common enough that we call problems using it *related rates* problems:

## Related Rates

Given a real-world/physical problem that involves:

- Some number quantities and measurements for them
- Some number of implicit relations between them
- Some information about the rate of changes of some of the quantities

and seeks information about the rate of changes of the other variables
solving the problem via the method of related rates involves:

1. Differentiating the relations between the quantities with respect to the variable of interest
2. Substituting in the known quantities
3. Solving the resulting equation(s) for the rate of change in question.

# Related Rates Example I

For a classic related rates problem: Suppose you have a cylindrical tank of water whose height is 10 meters and whose radius is 2 meters, with a hole at the bottom from which water drains out. At the moment the height of the water is halfway to the bottom, a hole at the bottom of the tank opens and you measure the speed at which the water escapes to be about $\frac{1}{2}\frac{m^3}{sec}$. At what rate will the height of the water decrease?
Identify information:

- Quantities: volume, height, radius. Radius is constant at 2 meters, volume changes w.r.t. height. Height is at 5 meters.
- Relations: $V = \pi r^2 h$.
- RoC information: $\frac{dV}{dt} = -\frac{1}{2}\frac{m^3}{sec}$.
- RoC in question: $\frac{dh}{dt}$.

# Related Rates Example I cont.

We then want to differentiate the relation (volume in terms of radius and height) with respect to time to relate the rates of change:

$$V = \pi r^2 h \tag{288}$$

$$\frac{d}{dt}V = \frac{d}{dt}(\pi r^2 h) \tag{289}$$

$$\frac{dV}{dt}\frac{d}{dV}V = \pi \frac{d}{dt}(r^2 h) \tag{290}$$

$$\frac{dV}{dt} \cdot 1 = \pi \cdot (\frac{d}{dt}(r^2) \cdot h + r^2 \cdot \frac{d}{dt}(h)) \tag{291}$$

$$\frac{dV}{dt} = \pi \cdot (\frac{dr}{dt} \cdot (\frac{d}{dr}r^2) \cdot h + r^2 \cdot \frac{dh}{dt}\frac{d}{dh}h) \tag{292}$$

$$\frac{dV}{dt} = \pi \cdot (0 \cdot _ + r^2 \frac{dh}{dt}) \quad \Longrightarrow \quad \frac{dV}{dt} = \pi r^2 \frac{dh}{dt} \tag{293}$$

$$-\frac{1}{2} = \pi \cdot 2^2 \frac{dh}{dt} \quad \Longrightarrow \quad \frac{dh}{dt} = -\frac{1}{8\pi}\frac{m}{sec} \tag{294}$$

So the height of the water is decreasing at a rate of $\frac{1}{8\pi}$ meters per second

# Related Rates Example II

Now suppose that the tank is instead an *inverted cone* with the same height and radius. What changes?

- The volume formula changes: $V = \frac{1}{3}\pi r^2 h$
- AND the radius also changes over time:

Giving us the *relation*:

$$V = \frac{1}{3}\pi r^2 h \tag{295}$$

$$V = \frac{1}{3}\pi \left(\frac{2}{10}h\right)^2 \cdot h \tag{296}$$

$$V = \frac{4\pi h^3}{300} = \frac{\pi h^3}{75} \tag{297}$$

# Related Rates Example II cont.

We would then differentiate this relation w.r.t. $t$ to get:

$$\frac{d}{dt}V = \frac{d}{dt}\frac{\pi h^3}{75} \tag{298}$$

$$\frac{dV}{dt} = \frac{\pi h^2}{25}\frac{dh}{dt} \tag{299}$$

$$\frac{dh}{dt} = \frac{25}{\pi h^2}\frac{dV}{dt} = \frac{25}{\pi \cdot 5^2} \cdot \frac{-1}{2} = \frac{-1}{2\pi}\frac{m}{s} \tag{300}$$

So now the height is decreasing at a rate of $\frac{1}{2\pi}$ meters per second.

# Related Rates Example III

For a non-tank draining problem, let's instead suppose there is a 10 meter long ladder propped up at an angle on a wall that makes a right angle with the ground. It's base is currently 6 meters away from the wall, and that base is sliding away from the wall at a rate of two meters per second. At that time, how high up along the wall is the ladder, and what is the rate of change of that height?

- Quantities: It is very helpful here to draw a picture!
  Our quantities are then the distance of the base of the ladder away from the wall $x(t)$ and how high along the wall the ladder is $y(t)$.
- Relations: The length of the ladder is constant $\implies x(t)^2 + y(t)^2 = 10^2$.
- RoC information: $\dfrac{dx}{dt} = 2\dfrac{m}{s}$.
- Information in question: value of $y(t)$ and $\dfrac{dy}{dt}$.

# Related Rates Example III cont.

We differentiate the relation w.r.t time:

$$x^2 + y^2 = 10^2 \tag{301}$$

$$\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}10^2 \tag{302}$$

$$\frac{d}{dt}(x^2) + \frac{d}{dt}(y^2) = 0 \tag{303}$$

$$\frac{dx}{dt}\frac{d}{dx}x^2 + \frac{dy}{dt}\frac{d}{dy}y^2 = 0 \tag{304}$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \tag{305}$$

$$2 \cdot 6 \cdot 2 + 2 \cdot y \cdot \frac{dy}{dt} = 0 \tag{306}$$

$$\frac{dy}{dt} = -\frac{12}{y}\frac{m}{sec} \tag{307}$$

# Related Rates Example III cont. 2

$$\frac{dy}{dt} = -\frac{12}{y}\frac{m}{sec} \tag{308}$$

So we still need to determine what $y(t)$ is when $x(t) = 6$. How?
Use the relation!

$$x(t)^2 + y(t)^2 = 10^2 \tag{309}$$
$$6^2 + y(t)^2 = 100 \tag{310}$$
$$y(t)^2 = 100 - 36 \tag{311}$$
$$y(t)^2 = 64 \quad \Longrightarrow \quad y(t) = \pm 8 \tag{312}$$

and distances must be positive so $y(t) = 8$ when $x(t) = 6$.
So:

$$\frac{dy}{dt} = -\frac{12}{y} = -\frac{12}{8} = -\frac{3}{2}\frac{m}{sec}. \tag{313}$$

The height of the ladder against the wall is therefore decreasing at a rate of $\frac{3}{2}$ meters per second.

# More to Come!

We'll also be spending next Tuesday going over more examples of how to use the method of related rates to solve different types of problems.
Generally however, all of these problems will rely on some kind of geometric input, e.g.:

- Area of square, rectangle, circle, ellipse, triangle
- Perimeter of the above (except ellipse)
- Angles (e.g. SOHCAHTOA trig)
- Volumes of cubes, rectangular prisms, pyramids, cones, spheres
- Surface areas of the cubes, rectangular prisms, spheres
- Pythagorean theorem
- Similar triangles

and thus having some way to commit these formulae to memory will be useful.

# Worksheet Time

Let's get into groups and work on the worksheet!

# End of Class

That’s all folks!

# Lecture 14 - L Hopitals Rule

# Lecture 14 - L Hopitals Rule

Garrett Credi

Summer 2026

# Updates

We have Exam 2 next week!
Practice exams and the review sheets are both out on Canvas, compiled lecture will be uploaded after class.
Solutions to the practice exam and review sheet will be uploaded Thursday night to give you all time to try out the problems before looking at the solutions.
If you're feeling worried about the exam, please feel free to come to my office hours or email me to set up a time to meet :)
Re: quizzes. I'll grind out quiz grading tonight and tomorrow. Let me know if you want me to give feedback on a specific quiz of yours earlier though :)

# Quiz Time

Let’s get into neat rows, spread out, and slay this quiz :)

# Limit Review I

Recall from the first two weeks of class:
we use *limits* to determine the values that functions approximate as their inputs approximate a given value.

They were very useful in examining *indeterminate forms*: expressions like $\frac{f(x)}{g(x)}$ around values $x = x_0$ where $f(x_0) = g(x_0) = 0$ or $\infty$.
Our tools included:

- Factoring out $(x - x_0)$
- Replacing the limit with the limit of a continuous function
- Addition/subtraction/product/quotient computations
- Squeeze theorem
- Etc.

# Limit Review II

We also defined derivatives in terms of limits:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{314}$$

But then quickly pivoted to constructing and working with the *synthetic* derivative rules, e.g.:

- Power Rule: $\frac{d}{dx} x^n = n x^{n-1}$
- Exponential/Logarithm rule: $\frac{d}{dx} e^x = e^x$, $\frac{d}{dx} \ln(x) = \frac{1}{x}$
- Trig Rules: $\frac{d}{dx} \sin(x) = \cos(x)$
- Sum/Difference rule:

$$\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$$

- Product Rule: $\frac{d}{dx} f(x)g(x) = \left(\frac{d}{dx} f(x)\right) \cdot g(x) + f(x) \cdot \left(\frac{d}{dx} g(x)\right)$
- Chain Rule: $\frac{d}{dx} g(f(x)) = \frac{dg}{du}\Big|_{u=f(x)} \cdot \frac{df}{dx}$.

# Derivative Review I

The importance of the derivative came from it's interpretation as *the best linear approximation*:

$$\lim_{h \to 0} \frac{|f(x) - (f(x_0) + f'(x_0)(x - x_0))|}{h} = 0 \tag{315}$$

or, in practical terms:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + O((x - x_0)^2) \tag{316}$$

We also saw the *quadratic approximation*:

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + O((x - x_0)^3) \tag{317}$$

# L'Hôpital's Rule I

Because of our ability to approximate functions via their linear/quadratic approximations, we gain another tool to put in our limit computing toolkit when we encounter indeterminate forms.

If $f(x)$ and $g(x)$ are two functions that are:

1. 0 when $x = x_0$
2. Differentiable at $x = x_0$

then, assuming $g'(x_0) \neq 0$, an application of squeeze theorem gives us:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f(x_0) + f'(x_0)(x - x_0)}{g(x_0) + g'(x_0)(x - x_0)} = \lim_{x \to x_0} \frac{f'(x_0)(x - x_0)}{g'(x_0)(x - x_0)} = \lim_{x \to x_0} \frac{f'(x_0)}{g'(x_0)} \tag{318}$$

We can also work around $g'(x_0)$ being zero by instead reducing

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \tag{319}$$

# L'Hôpital's Rule II

**(Mini) L'Hôpital's Rule**

If $f(x)$, $g(x)$ are two functions that are:

1. Zero at $x = x_0$ (i.e. $f(x_0) = g(x_0) = 0$)
2. Differentiable at $x = x_0$

then:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \tag{320}$$

In other words, we can compute limits of indeterminate forms by *differentiating the two functions* to kill off any obstructions to continuity.

# L'Hôpital's Rule III

This comes from a generalization of both linear and quadratic approximation, arguably the most important concept taught in Calculus II:

**Taylor's Theorem**

For $f(x)$ a function whose higher derivatives $\frac{d^n f}{dx^n}\,|_{x=x_0}$ all exist one can form the *Taylor Polynomial* of $f(x)$ at $x = x_0$ defined as:

$$T_f(x) := \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \tag{321}$$

and as $x \to x_0$ the error term $E(x) = |f(x) - T_f(x)|$ approaches zero *extremely rapidly*. (And in most nice cases it turns out that $f(x) = T_f(x)$ *literally* on a small interval containing $x_0$).

(this is not material you are expected to remember for this class)

# L'Hôpital's Rule IV

L'Hôpital's Rule is then just what happens when you compute $\lim_{x \to x_0} \frac{f(x)}{g(x)}$ by replacing it with $\lim_{x \to x_0} \frac{T_f(x)}{T_g(x)}$ using the Squeeze Theorem. E.g.:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{6}(x - x_0)^3 + \ldots}{g(x_0) + g'(x_0)(x - x_0) + \frac{g''(x_0)}{2}(x - x_0)^2 + \frac{g'''(x_0)}{6}(x - x_0)^3 + \ldots} \tag{322}$$

$$= \lim_{x \to x_0} \frac{f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{6}(x - x_0)^3 + \ldots}{g'(x_0)(x - x_0) + \frac{g''(x_0)}{2}(x - x_0)^2 + \frac{g'''(x_0)}{6}(x - x_0)^3 + \ldots} \tag{323}$$

$$= \lim_{x \to x_0} \frac{f'(x_0) + \frac{f''(x_0)}{2}(x - x_0) + \frac{f'''(x_0)}{6}(x - x_0)^2 + \ldots}{g'(x_0) + \frac{g''(x_0)}{2}(x - x_0) + \frac{g'''(x_0)}{6}(x - x_0)^2 + \ldots} \tag{324}$$

# L'Hôpital's Rule V

What if $\lim_{x \to x_0} \frac{f'(x)}{g'(x)}$ is another $\frac{0}{0}$ indeterminate form? e.g. $f'(x_0) = g'(x_0) = 0$?

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x_0) + \frac{f''(x_0)}{2}(x - x_0) + \frac{f'''(x_0)}{6}(x - x_0)^2 + \ldots}{g'(x_0) + \frac{g''(x_0)}{2}(x - x_0) + \frac{g'''(x_0)}{6}(x - x_0)^2 + \ldots} \tag{325}$$

$$= \lim_{x \to x_0} \frac{\frac{f''(x_0)}{2}(x - x_0) + \frac{f'''(x_0)}{6}(x - x_0)^2 + \ldots}{\frac{g''(x_0)}{2}(x - x_0) + \frac{g'''(x_0)}{6}(x - x_0)^2 + \ldots} \tag{326}$$

$$= \lim_{x \to x_0} f''(x_0) + \frac{f'''(x_0)}{3}(x - x_0) + \ldots g''(x_0) + \frac{g'''(x_0)}{3}(x - x_0) + \ldots \tag{327}$$

In other words, we just L'Hôpital it again!

# L'Hôpital's Rule VI

This gives us a general procedure to compute limits of the form $\lim_{x \to x_0} \frac{f(x)}{g(x)}$:

## L'Hôpital's Process for Computing Limits of the Indeterminate Form $\frac{0}{0}$

To compute $\lim_{x \to x_0} \frac{f(x)}{g(x)}$:

1. Compute $f(x_0)$ and $g(x_0)$.
2. If $f(x_0) \neq 0$ and $g(x_0) \neq 0$, then $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{f(x_0)}{g(x_0)}$.
3. If $f(x_0) \neq 0$ but $g(x_0) = 0$ then we're ending up with something like $\frac{N}{0}$ so: $\lim_{x \to x_0} \frac{f(x)}{g(x)}$ DNE
4. If $f(x_0) = g(x_0) = 0$ and $f(x)$, $g(x)$ are differentiable at $x = x_0$ , replace $f(x)$ with $f'(x)$ and $g(x)$ with $g'(x)$ and restart step 1.

# L'Hôpital's Rule VII

What about other indeterminate forms like:

$$\frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad 0^0, \quad \infty^0, \quad 1^\infty, \quad \infty - \infty?$$

If L'Hôpital's rule helps us compute limits of indeterminate forms like $\frac{0}{0}$ why not just try and force those above forms into something related enough and that looks like $\frac{0}{0}$!

E.g. if we had $\lim_{x \to 0} \frac{\cot(x)}{\ln(x^2)}$ we see this is of the form $\frac{\infty}{\infty}$ but we can change it to:

$$\lim_{x \to 0} \frac{\cot(x)}{\ln(x^2)} = \left( \lim_{x \to 0} \frac{\frac{1}{\ln(x^2)}}{\frac{1}{\cot(x)}} \right)$$

which is of the form $\frac{0}{0}$.

# L'Hôpital's Rule Examples I

We can then compute:

$$\lim_{x\to 0}\frac{\cot(x)}{\ln(x^2)} = \lim_{x\to 0}\frac{\frac{1}{\ln(x^2)}}{\tan(x)} \tag{328}$$

$$= \lim_{x\to 0}\frac{\frac{d}{dx}(\ln(x^2))^{-1}}{\frac{d}{dx}\tan(x)} \tag{329}$$

$$= \lim_{x\to 0}\frac{2x\cdot\frac{1}{(x^2)}\cdot -1\cdot(\ln(x^2))^{-2}}{\sec(x)^2} \tag{330}$$

$$= \lim_{x\to 0}\frac{\frac{-2\ln(x^2)}{x}}{\sec(x)^2} \tag{331}$$

since $\sec(0)^2 = 1$, and since the numerator is an indeterminate form of type $\frac{\infty}{0}$, we can conclude this limit DNE.

# L'Hôpital's Rule Examples II

If you're a fan of Mean Girls, you might be interested in computing:

$$\lim_{x \to 0} \frac{\ln(1-x) - \sin(x)}{1 - \cos(x)^2} \tag{332}$$

We check @ $x = 0$, we'd get: $\frac{\ln(1-0) - \sin(0)}{1 - \cos(0)^2} = \frac{\ln(1) - 0}{1 - 1^2} = \frac{0}{0}$ so this is already in a form amenable to L'Hôpital's Rule!

$$\lim_{x \to 0} \frac{\ln(1-x) - \sin(x)}{1 - \cos(x)^2} = \lim_{x \to 0} \frac{\frac{d}{dx}(\ln(1-x) - \sin(x))}{\frac{d}{dx}(1 - \cos(x)^2)} \tag{333}$$

$$= \lim_{x \to 0} \frac{\frac{-1}{1-x} - \cos(x)}{-2\cos(x) \cdot -\sin(x)} \tag{334}$$

Which at 0 is $\frac{\frac{-1}{1-0} - \cos(0)}{2\cos(0)\sin(0)} = \frac{-2}{0}$ so the limit does not exist!

# L'Hôpital's Rule Examples III

Another one:

$$\lim_{x \to 0} x^x \tag{335}$$

Indeterminate form of type $0^0$. What can we use to turn exponentiation into a product/quotient of things? Logarithms!

$$\ln\left(\lim_{x \to 0} x^x\right) = \lim_{x \to 0} \ln(x^x) = \lim_{x \to 0} x \ln(x) \tag{336}$$

$$= \lim_{x \to 0} \frac{\ln(x)}{\frac{1}{x}} = \lim_{x \to 0} \frac{\frac{d}{dx}\ln(x)}{\frac{d}{dx}\frac{1}{x}} \tag{337}$$

$$= \lim_{x \to 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \to 0} -\frac{x^2}{x} = 0 \tag{338}$$

So $\ln(\lim_{x \to 0} x^x) = 0$, which means $\lim_{x \to 0} x^x = e^0 = 1$.

# L'Hôpital's Rule Cheat Method!

In the previous example we did literal L'Hôpital's rule on an indeterminate form of type $\frac{\infty}{\infty}$. Why is this valid?

$$\lim_{x\to x_0}\frac{f(x)}{g(x)} = \lim_{x\to x_0}\frac{\frac{1}{g(x)}}{\frac{1}{f(x)}} \tag{339}$$

$$= \lim_{x\to x_0}\frac{\frac{d}{dx}\frac{1}{g(x)}}{\frac{d}{dx}\frac{1}{f(x)}} \tag{340}$$

$$= \lim_{x\to x_0}\frac{\frac{-g'(x)}{g(x)^2}}{\frac{-f'(x)}{f(x)^2}} = \left(\lim_{x\to x_0}\frac{g'(x)}{f'(x)}\right)\cdot\left(\lim_{x\to x_0}\frac{f(x)}{g(x)}\right)^2 \tag{341}$$

from which you can solve that, for $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = \infty$

$$\lim_{x\to x_0}\frac{f(x)}{g(x)} = \lim_{x\to x_0}\frac{f'(x)}{g'(x)} \tag{342}$$

# Superpowered L'Hôpital's Rule

## L'Hôpital's Rule

For $f(x)$, $g(x)$ two functions where *either*:

1. $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$ and $f(x), g(x)$ are differentiable at $x = x_0$
2. $\lim_{x\to x_0} |f(x)| = \lim_{x\to x_0} |g(x)| = \infty$ and $\frac{1}{f(x)}, \frac{1}{g(x)}$ are differentiable at $x = x_0$.

Then we have that:

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \lim_{x\to x_0} \frac{f'(x)}{g'(x)} \tag{343}$$

Then to compute the limit of *any* indeterminate form, we need only equate it to some indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and use L'Hôpital's Rule to compute it.

# L'Hôpital's Rule Examples IV

Let's consider $\lim_{x\to 0}\frac{\sin(x)}{\cos(x)}$ as well as the following "computation":

$$\lim_{x\to x_0}\frac{\sin(x)}{\cos(x)} = \lim_{x\to x_0}\frac{\frac{d}{dx}\sin(x)}{\frac{d}{dx}\cos(x)} \tag{344}$$

$$= \lim_{x\to 0}\frac{\cos(x)}{-\sin(x)} \tag{345}$$

DNE since $\cos(0) = 1$ while $-\sin(0) = 0$ so we end up with $\frac{1}{0}$ so DNE. Thus $\tan(0)$ does not exist.

WE DIDN'T ACTUALLY CHECK IF $\frac{\sin(x)}{\cos(x)}$ WAS AN INDETERMINATE FORM AT $x = 0$!

$\sin(0) = 0$ and $\cos(0) = 1$ so $\frac{\sin(x)}{\cos(x)}$ is *continuous* at $x = 0$ so

$\lim_{x\to 0}\frac{\sin(x)}{\cos(x)} = \frac{\sin(0)}{\cos(0)} = \frac{0}{1} = 0 = \tan(0)$.

# L'Hôpital's Rule Examples V

Finally, let's consider

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x \tag{346}$$

(which actually matters since its the "rate of continuously compounding interest")
What type of indeterminate form? $\lim_{x\to\infty} 1+\frac{1}{x} = 1$, so $1^\infty$.
How do we turn exponentiation into a product/quotient? Log!

$$\ln\left(\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x\right) = \lim_{x\to\infty}\ln\left(\left(1+\frac{1}{x}\right)^x\right) \tag{347}$$

$$= \lim_{x\to\infty} x\ln\left(1+\frac{1}{x}\right) = \lim_{x\to\infty}\frac{\ln(1+\frac{1}{x})}{\frac{1}{x}} \tag{348}$$

$$= \lim_{x\to\infty}\frac{-\frac{1}{x^2}\cdot\frac{1}{1+\frac{1}{x}}}{-\frac{1}{x^2}} = \lim_{x\to\infty}\frac{x}{x+1} = 1 \tag{349}$$

So $\ln\left(\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x\right) = 1$ and therefore $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e$.

# Worksheet Time!

Let’s get into groups and work on the worksheet :)

# End of Class

That’s all folks :)

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